

SGA 3, Exposé XIV
Regular Elements: Continuation
Applications to Algebraic Groups
English Reconstruction

A. Grothendieck
with an Appendix by J.-P. Serre

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1 Construction of Cartan subgroups and maximal tori for a smooth algebraic group

Theorem 1.1. *Let G be a smooth algebraic group over a field k . Then G admits a maximal torus T , and hence a Cartan subgroup $C = \text{Centr}_G(T)$.*

By XII 3.2, finding a maximal torus T of G is equivalent to finding a Cartan subgroup C of G . Moreover, since the maximal tori of G are those of G^0 , we may suppose that G is connected. We distinguish two cases.

1°) *The field k is finite.* Let $\mathcal{T}$ be the scheme of maximal tori of G (XII 1.10); it is smooth over k . The group G acts on $\mathcal{T}$ through inner automorphisms, and the conjugacy theorem (XII 6.6(a)) says that any two $\bar{k}$ -rational points of $\mathcal{T}_{\bar{k}}$ are conjugate under $G_{\bar{k}}(\bar{k})$. Since $\mathcal{T}$ is smooth over k , $\mathcal{T}_{\bar{k}}$ is smooth over $\bar{k}$; it follows that $\mathcal{T}_{\bar{k}}$ is isomorphic to

$$G_{\bar{k}}/N,$$

where N is the stabilizer of an element of $\mathcal{T}_{\bar{k}}(\bar{k})$, that is, the normalizer of a maximal torus $\bar{T}$ of $G_{\bar{k}}$. Consequently $\mathcal{T}$ is a homogeneous space under the action of G . A well-known theorem of Lang (*Amer. J. Math.* **78** (1956), 555–563) says that every homogeneous space under a smooth connected algebraic group over a finite field k has a rational point. In particular, $\mathcal{T}$ has a rational point; equivalently, G has a maximal torus T .

2°) *The field k is infinite.* We shall use the following lemma.

Lemma 1.2. *Let G be a smooth algebraic group over a field k . Then G admits a subgroup of type (C) (XIII 6.2).*

By XIII 6.3, this is equivalent to saying that $\mathfrak{g}$ contains a Cartan subalgebra $\mathfrak{d}$. When k is infinite this is immediate: $\mathfrak{g}$ contains a regular element a , and one takes

$$\mathfrak{d} = \text{Nil}(a, \mathfrak{g}).$$

The case of finite k is treated exactly as in the proof of 1°) above, but it requires first constructing the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$ and proving that this scheme is smooth over k ; this will be done below in 2.16.

To prove 1.1 in case 2°), which is the case now under consideration, it is therefore enough to know 1.2 when k is infinite. We also record the following result for later use.

Lemma 1.3. *Let G be a smooth connected affine algebraic group whose reductive center (XII 4.1 and 4.4) is reduced to the identity group. Then G is nilpotent if and only if its Lie algebra $\mathfrak{g}$ is nilpotent.*

This is contained in XII 4.9.

We can now give a procedure for constructing Cartan subgroups of G . The same procedure applies when k is finite, provided Lemma 1.2 is granted in that case. Suppose first that G is affine. We argue by induction on $n = \dim G$, the assertion being trivial when $n = 0$. Suppose therefore that $n > 0$, and that the assertion has been proved in dimensions $n' < n$. Let Z be the reductive center of G , and let

$$u: G \longrightarrow G' = G/Z$$

be the canonical homomorphism. By XII 4.7(c), the rule

$$C' \longmapsto C = u^{-1}(C')$$

⁰Version xy of 1 December 2008.

is a bijection from the Cartan subgroups of G' to those of G . Replacing G by G' , we may thus suppose that the reductive center Z of G is reduced to the identity element (the same is true for the reductive center of G' , by XII 4.7(b)).

By Lemma 1.2,¹ G admits a subgroup D of type (C). Over the algebraic closure $\bar{k}$ of k , the group $D_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$ (XIII 6.6(b)); hence every Cartan subgroup of D is a Cartan subgroup of G (XIII 2.8(a)). We are therefore reduced to finding a Cartan subgroup of D .

If $\dim D = \dim G$, that is, if $D = G$, then the Lie algebra of G is a Cartan subalgebra of itself and is therefore nilpotent. Lemma 1.3 then shows that G is nilpotent, so G is a Cartan subgroup of itself. If $\dim D < \dim G$, the induction hypothesis gives a Cartan subgroup of D , which is consequently a Cartan subgroup of G . This proves the assertion when G is affine.

In the general case, let Z be the center of G . Then $G/Z = G'$ is affine (XII 6.1), and the inverse image in G of every Cartan subgroup C' of G' is a Cartan subgroup of G (XII 6.6(e)). We are thus reduced to finding a Cartan subgroup of the smooth affine algebraic group G' , which is the case already treated.

Corollary 1.4. *Let G be a smooth group scheme of finite type over an Artinian scheme S . Then G admits a maximal torus T , and hence a Cartan subgroup $C = \text{Centr}_G(T)$. Every torus in G is contained in a maximal torus.*

Indeed, we may suppose that S is local, with residue field k . By 1.1, the group

$$G_0 = G \times_S \text{Spec}(k)$$

admits a maximal torus T_0 . By IX 3.6 bis and X 2.3, T_0 comes from a torus T of G , which is plainly maximal. The last assertion follows by applying the preceding result to the centralizer of the given torus, which is smooth over k (XI 2.4).

Remarks 1.5. (a) We shall give below, in 3.20, 3.21, and Exposé XV, variants of 1.4 in which S is not assumed Artinian.

(b) I do not know whether every algebraic group G , not necessarily smooth, over a field k admits a maximal torus. The question arises only in characteristic $p > 0$. Applying 1.1 to a smooth quotient group of the form $G' = G/I$, where I is a suitable radicial subgroup of G (for example, the kernel of a sufficiently high power of the Frobenius homomorphism), and then taking in G the inverse image of a maximal torus of G' , reduces the question to the case in which $(G_{\bar{k}})_{\text{red}}$ is a torus; here, as always, $\bar{k}$ denotes the algebraic closure of k . The answer is easily seen to be affirmative when G is commutative, or more generally nilpotent: in that case G has a unique maximal torus, which may for example be constructed by descent from the maximal torus of $G_{\bar{k}}$.^{*}

(c) When G is affine and either k is perfect or G is solvable, 1.1 was already known, by a theorem of Rosenlicht; his proof is quite different from the one given here.

(d) When k is infinite, 1.1 follows from the much more precise result, proved below in 6.1, that the scheme $\mathcal{T}$ of maximal tori of G is a rational variety. The method essentially combines the proof of 1.1 with an explicit description of the structure of the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$. To obtain the desired result, we must first extend to an arbitrary base prescheme certain results of XIII 4–6 (this is the purpose of the next two sections),

¹The source prints “1.3” here; the cited existence result is Lemma 1.2.

^{*}M. Raynaud gave a negative answer to the question raised here; see XVII, Example 5.9(c).

and then refine the preceding construction proving 1.1, using the fact that every Cartan subgroup of G is contained in one and only one subgroup of type (C) in G (Nos. 4 and 5).

2 Lie algebras over an arbitrary prescheme: regular sections and Cartan subalgebras

Lemma 2.1. *Let A be a ring, let $\mathfrak{d}$ be a Lie algebra over A , and for each $s \in \operatorname{Spec}(A)$ let*

$$\mathfrak{d}(s) = \mathfrak{d} \otimes_A k(s)$$

be the corresponding Lie algebra over the residue field $k(s)$. Suppose that the A -module $\mathfrak{d}$ is of finite presentation. The following conditions are equivalent:

- (i) *for every $s \in \operatorname{Spec}(A)$, the Lie algebra $\mathfrak{d}(s)$ is nilpotent;*
- (ii) *for every $x \in \mathfrak{d}$, the endomorphism $\operatorname{ad}(x)$ is nilpotent;*
- (iii) *there is an integer $N \geq 0$ such that, for every sequence $x_1, \dots, x_N$ of elements of $\mathfrak{d}$, one has*

$$\operatorname{ad}(x_1) \operatorname{ad}(x_2) \cdots \operatorname{ad}(x_N) = 0.$$

When A is a field, the equivalence of (i) and (iii) is the definition of nilpotence, while the equivalence of (ii) and (iii) is the familiar consequence of Engel's theorem (Bourbaki, *Groupes et algèbres de Lie*, Chapter I, §4, No. 2). In the general case, (iii) plainly implies (ii), and (ii) implies (i) by the preceding result and the fact that (ii) is preserved under passage to quotients and under localization. It remains to prove that (i) implies (iii).

Suppose first that A is local Artinian, with maximal ideal $\mathfrak{m}$. Choose $n > 0$ such that $\mathfrak{m}^n = 0$, and choose N such that condition (iii) holds for

$$\mathfrak{d}(s) = \mathfrak{d} \otimes_A A/\mathfrak{m}.$$

Taking $N' = nN$, one sees at once that N' satisfies (iii) for $\mathfrak{d}$.

If A is Noetherian, there are finitely many points $s_i \in \operatorname{Spec}(A)$ and ideals of definition $\mathfrak{q}_i \subset A_{s_i}$ such that the natural map

$$\mathfrak{d} \longrightarrow \prod_i (\mathfrak{d} \otimes_A A_i), \quad A_i = A_{s_i}/\mathfrak{q}_i,$$

is injective. By the preceding case, for each i there is an integer N_i satisfying (iii) for the Lie algebra $\mathfrak{d} \otimes_A A_i$. Taking N to be the largest of the N_i , condition (iii) follows for $\mathfrak{d}$. Finally, the general case reduces to the Noetherian case by the usual procedure described in EGA IV 8.

Definition 2.2. Let S be a prescheme and let $\mathfrak{d}$ be a quasi-coherent Lie algebra over S whose underlying module is of finite presentation. We say that $\mathfrak{d}$ is *locally nilpotent* if, for every $s \in S$, the Lie algebra $\mathfrak{d}(s)$ over $k(s)$ is nilpotent. We say that $\mathfrak{d}$ is *strictly locally nilpotent* if it is locally free and, on every open subset U of S over which it has constant rank r , its Killing polynomial reduces to

$$P_{\mathfrak{d}}(t) = t^r.$$

Of course, when $\mathfrak{d}$ is a Lie algebra over S that is locally free as a module, its Killing polynomial is defined as a polynomial

$$P_{\mathfrak{d}} \in A[t], \quad A = \Gamma(\mathrm{Sym}_{\mathcal{O}_S}(\mathfrak{d})) \simeq \Gamma(W(\mathfrak{d})),$$

where A is the ring of sections of the structure sheaf of the vector bundle $W(\mathfrak{d})$ defined by $\mathfrak{d}$.

The two notions introduced in 2.2 are plainly stable under base change and local for the fpqc topology. We record:

Proposition 2.3. *If $\mathfrak{d}$ is strictly locally nilpotent, then it is locally nilpotent. The converse holds when $\mathfrak{d}$ is locally free and S is reduced.*

The proof is immediate.

Definition 2.4. Let S be a prescheme, let $\mathfrak{g}$ be a Lie algebra over S whose underlying module is locally free of finite type, and let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$. We call $\mathfrak{d}$ a *Cartan subalgebra* if it satisfies the following conditions:

- (i) $\mathfrak{d}$ is locally a direct-factor submodule (and hence is itself locally free of finite type);
- (ii) for every $s \in S$, the Lie algebra $\mathfrak{d}(s)$ is a Cartan subalgebra of $\mathfrak{g}(s)$.

Definition 2.5. Let S and $\mathfrak{g}$ be as in 2.4. A section a of $\mathfrak{g}$ is called *quasi-regular* if, for every $s \in S$, the element $a(s) \in \mathfrak{g}(s)$ is regular in the Lie algebra $\mathfrak{g}(s)$ over $k(s)$. We call a a *regular section* if it is quasi-regular and is contained in a Cartan subalgebra of $\mathfrak{g}$.

The notions in 2.4 and 2.5 are again stable under base change and local for the fpqc topology. Only the last assertion, and only for the notion of a regular section, requires proof. It follows from the fact that the Cartan subalgebra containing a given regular section is uniquely determined. More precisely:

Proposition 2.6. *Let S and $\mathfrak{g}$ be as in 2.4, and let a be a quasi-regular section of $\mathfrak{g}$. There is at most one Cartan subalgebra of $\mathfrak{g}$ containing a . Such a Cartan subalgebra exists—that is, a is a regular section—if and only if a satisfies the following condition:*

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g})$$

is locally a direct-factor submodule of $\mathfrak{g}$, and $\mathrm{ad}(a)$ induces an automorphism of $\mathfrak{g}/\mathfrak{d}$. In that case $\mathfrak{d}$ is the unique Cartan subalgebra of $\mathfrak{g}$ containing a .

Suppose first that a is contained in a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$. Then $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is bijective on every fiber. Since $\mathfrak{g}/\mathfrak{d}$ is locally free of finite type, it follows that this endomorphism is an automorphism of $\mathfrak{g}/\mathfrak{d}$. On the other hand, by 2.1 the endomorphism $\mathrm{ad}(a)_{\mathfrak{d}}$ is locally nilpotent. Hence

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g}),$$

which proves both the uniqueness of $\mathfrak{d}$ and the necessity of the stated regularity criterion.

For sufficiency, observe that the hypothesis on a implies that the formation of $\mathrm{Nil}(a, \mathfrak{g})$ commutes with base extension, and in particular with passage to fibers. Thus the fibers $\mathfrak{d}(s)$ of $\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g})$ are Cartan subalgebras of the $\mathfrak{g}(s)$. Moreover, $\mathfrak{d}$ is a Lie subalgebra of $\mathfrak{g}$ by XIII 4.1. It is therefore a Cartan subalgebra.

Corollary 2.7. *Under the hypotheses of 2.4, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$, let a be a section of $\mathfrak{d}$ that is regular in $\mathfrak{g}$, and let u be an automorphism of $\mathfrak{g}$. Then u preserves $\mathfrak{d}$ if and only if $u(a)$ is a section of $\mathfrak{d}$.*

Indeed, by transport of structure $u(a)$ is then a regular section of $\mathfrak{g}$, contained in both Cartan subalgebras $\mathfrak{d}$ and $u(\mathfrak{d})$; Proposition 2.6 therefore shows that these two subalgebras are equal.

Corollary 2.8. *Under the hypotheses of 2.4, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. For every $s \in S$ such that $k(s)$ is infinite, there exist an open neighborhood V of s and a regular section a of $\mathfrak{g}$ over V such that*

$$\mathfrak{d}|_V = \text{Nil}(a, \mathfrak{g}|_V);$$

equivalently, a is a section of $\mathfrak{d}|_V$.

Indeed, since $k(s)$ is infinite, there is a regular element a_0 of $\mathfrak{g}(s)$ contained in $\mathfrak{d}(s)$. After replacing S by an open neighborhood U of s , we may extend a_0 to a section a of $\mathfrak{d}$ over U . The endomorphism induced by $\text{ad}(a)$ on the fiber

$$\mathfrak{g}(s)/\mathfrak{d}(s)$$

is an automorphism. Shrinking U to a smaller open neighborhood V , we may therefore suppose that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is an automorphism. It follows that a is a quasi-regular section of $\mathfrak{g}|_V$ satisfying the required condition.

Let $\mathfrak{g}$ be as in 2.4. Inspection of its Killing polynomial immediately shows that the function

$$s \longmapsto \text{the nilpotent rank of } \mathfrak{g}(s)$$

on S is upper semicontinuous. We shall be especially interested in the case in which this function is continuous, that is, locally constant. Here are several variants of this property.

Proposition 2.9. *Let $S, \mathfrak{g}$ be as in 2.4. Consider the following conditions:*

- (C₀) *The nilpotent rank of $\mathfrak{g}(s)$, for $s \in S$, is a locally constant function of s .*
- (C₁) *Locally for the fpqc topology, there exists a Cartan subalgebra of $\mathfrak{g}$.*
- (C'₁) *The same as (C₁), with “fpqc topology” replaced by “étale topology”.*
- (C₂) *Condition (C₀) holds, and for every S' over S , every quasi-regular section of*

$$\mathfrak{g}_{S'} = \mathfrak{g} \otimes_S S'$$

is regular.

- (C₃) *Every $s \in S$ has an open neighborhood V on which the Killing polynomial of $\mathfrak{g}$ has the form*

$$t^r (t^{n-r} + c_1 t^{n-r-1} + \cdots + c_{n-r}),$$

where, for every $s \in V$,

$$c_{n-r}(s) \in \text{Sym}(\mathfrak{g}(s)^\vee)$$

is nonzero.

With this notation, one has the implications

$$(C_3) \implies (C_2) \implies (C'_1) \implies (C_1) \implies (C_0),$$

and, when S is reduced, the five conditions above are equivalent.

Notice also that these conditions are manifestly stable under base change and are local for the fpqc topology.

The implications

$$(C'_1) \implies (C_1) \implies (C_0)$$

are trivial, and the implication $(C_0) \implies (C_3)$ when S is reduced is immediate. We note moreover:

Corollary 2.10. *Suppose that condition (C_0) holds. Let U be the set of elements of $W(\mathfrak{g})$ that are regular in their fiber. Then U is open. In particular, for every section a of $\mathfrak{g}$ over S , the set V of $s \in S$ for which $a(s) \in \mathfrak{g}(s)$ is regular is open.*

Indeed, the first assertion follows from the second, applied to $\mathfrak{g}_{S'}$ for every base change S' over S . For the second, we may suppose that S is reduced; then (C_3) holds, and it is enough to inspect the Killing polynomial of a in $\mathfrak{g}$.

The implication $(C_2) \implies (C'_1)$ now follows easily from part b) of the following more precise statement.

Corollary 2.11. *Suppose that condition (C_2) holds. Then:*

a) *For every $s \in S$ and every Cartan subalgebra $\mathfrak{d}_0$ of $\mathfrak{g}(s)$ such that $\mathfrak{d}_0$ contains a regular element of $\mathfrak{g}(s)$ (a condition automatically satisfied when $k(s)$ is infinite), there exist an open neighborhood V of s and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$ whose fiber at s is $\mathfrak{d}_0$. If S_1 is a subscheme of S containing s , and if $\mathfrak{d}_0$ has already been extended to a Cartan subalgebra $\mathfrak{d}_1$ of $\mathfrak{g} \otimes_S S_1$, then one can find an open neighborhood V of s in S and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$ such that*

$$\mathfrak{d} \otimes_V (S_1 \cap V) = \mathfrak{d}_1|_{S_1 \cap V}.$$

b) *For every $s \in S$ such that $\mathfrak{g}(s)$ contains a regular element (a condition automatically satisfied when $k(s)$ is infinite), there exist an open neighborhood V of s and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$.*

Part b) follows from part a) by taking

$$\mathfrak{d}_0 = \text{Nil}(a_0, \mathfrak{g}(s)),$$

where a_0 is a regular element of $\mathfrak{g}(s)$. To prove part a), in its second formulation, choose a regular element a_0 of $\mathfrak{g}(s)$ contained in $\mathfrak{d}_0$. Extend it near s in S_1 to a section of $\mathfrak{d}_1$, and extend that section in turn near s to a section of $\mathfrak{g}$. By 2.10 this section is quasi-regular on an open neighborhood V of s , hence regular by (C_2) . Therefore, by 2.6,

$$\mathfrak{d} = \text{Nil}(a, \mathfrak{g}|_V)$$

has the required properties.

Remark 2.12. I do not know whether the assertions of 2.11 a) and b) remain valid without the hypothesis that regular points exist, when $k(s)$ is finite.

It remains to prove

$$(C_3) \implies (C_2).$$

We also record the following equivalent form of (C_3) :

(C'_3) Condition (C_0) holds, that is, the nilpotent rank of $\mathfrak{g}(s)$, for $s \in S$, is locally constant; and on every open subset V of S on which this rank has the value r , the Killing polynomial of $\mathfrak{g}|_V$ is divisible by t^r .

We must show that this condition implies that every quasi-regular section a of $\mathfrak{g}$ is regular. In view of 2.6, this follows from the implication (iv) $\implies$ (iii) in the following lemma, applied to the endomorphism $\text{ad}(a)$ of $\mathfrak{g}$.

Lemma 2.13. *Let A be a ring, let M be a projective A -module of finite type, and let u be an endomorphism of M . The following conditions are equivalent:*

- (i) *M is the direct sum of two u -stable submodules M' and M'' , such that $u|_{M'}$ is nilpotent and $u|_{M''}$ is an automorphism of M'' .*
- (ii) *There exists an integer $n > 0$ such that*

$$\text{Im } u^n + \text{Ker } u^n = M.$$

- (iii) *The nilspace*

$$\text{Nil}(u) = \bigcup_{n>0} \text{Ker } u^n$$

is a direct factor of M , and

$$M = \text{Nil}(u) + u(M).$$

These conditions are implied by the following one, and are equivalent to it when A is reduced:

- (iv) *Locally on $\text{Spec}(A)$, for the Zariski topology, the characteristic polynomial $P_u(t)$ of u can be written in the form*

$$t^r (t^{n-r} + c_1 t^{n-r-1} + \cdots + c_{n-r}),$$

where c_{n-r} is invertible.

The equivalence of (i), (ii), and (iii) is immediate and is recalled for reference. When A is reduced, (i) implies (iv) because a nilpotent endomorphism of a projective module of rank r then has characteristic polynomial t^r . On the other hand, in all cases the constant term of the characteristic polynomial of an automorphism of a projective module of finite type is, up to sign, the determinant of u , locally on $\text{Spec}(A)$; it is therefore an invertible element of A .

Finally, to prove (iv) $\implies$ (i), regard M as a module over the polynomial ring $A[t]$, with t acting as u . The familiar identity

$$P(u) = 0$$

shows that M is annihilated by $PA[t]$, and hence may be regarded as a module over $A[t]/PA[t]$. Write

$$P = t^r Q,$$

where the constant term of Q is invertible. Then

$$PA[t] = t^r A[t] \cap QA[t],$$

so that $A[t]/PA[t]$ decomposes as the product of the rings

$$A[t]/t^r A[t] \quad \text{and} \quad A[t]/QA[t].$$

This gives a corresponding decomposition of M into the direct sum of two $A[t]$ -modules, or equivalently two u -stable A -submodules M' and M'' ; this is precisely the decomposition of (i).

This completes the proof of 2.13, and hence that of 2.9.

Corollary 2.14. *When condition (C_3) of 2.9 holds, the Cartan subalgebras of $\mathfrak{g}$ are strictly nilpotent (2.2).*

The proof is immediate.

Remark 2.15. a) One can prove a converse to 2.14: condition (C_3) is equivalent to the assertion that, for every S' over S , every quasi-regular section of $\mathfrak{g}_{S'}$ is regular and every Cartan subalgebra of $\mathfrak{g}_{S'}$ is strictly nilpotent; equivalently, every quasi-regular section of $\mathfrak{g}_{S'}$ is contained in a strictly nilpotent Cartan subalgebra of $\mathfrak{g}_{S'}$.

b) Unlike the other conditions (C_0) through (C_2) , condition (C_3) is of “infinitesimal nature.” More precisely, when S is locally noetherian, $\mathfrak{g}$ satisfies (C_3) if and only if, for every base change $S' \rightarrow S$ with S' local Artinian, $\mathfrak{g}_{S'}$ satisfies the same condition. If desired, one may take S' to be the spectrum of an Artinian quotient of a local ring of S . Similarly, when (C_0) holds, the condition for a section of $\mathfrak{g}$ to be regular is of infinitesimal nature.

c) When $\mathfrak{g}$ is the Lie algebra of a smooth group prescheme of finite presentation over S , we shall see that the conditions

$$(C_0), \quad (C_1), \quad (C'_1), \quad (C_2)$$

on $\mathfrak{g}$ are equivalent (5.2 a). I do not know what happens in general, except that even when S is local Artinian, (C_0) does not imply (C_1) . However, even when $\mathfrak{g}$ comes from a group G and S is local Artinian, it is not true in general that (C_2) implies (C_3) , since $\mathfrak{g}$ may be nilpotent without being strictly nilpotent.

An example is obtained from a smooth affine group scheme G over the spectrum S of a discrete valuation ring, such that the Lie algebra of the generic fiber is not nilpotent—for example, the generic fiber may be an adjoint semisimple group—whereas that of the special fiber is nilpotent—for example, the special fiber may be a vector group. Then, for sufficiently large n , the Lie algebra of

$$G_n = G \times_S S_n$$

is not strictly nilpotent, although it is nilpotent.

We continue under the hypotheses of 2.4. Let

$$\mathcal{D}: ((\text{Sch})/S)^\circ \longrightarrow \text{Sets}$$

be the functor defined by

$$\mathcal{D}(S') = \{\text{Cartan subalgebras of } \mathfrak{g}_{S'}\}.$$

Introduce also the functor

$$X(S') = \left\{ (\mathfrak{d}, a) \left| \begin{array}{l} \mathfrak{d} \text{ is a Cartan subalgebra of } \mathfrak{g}_{S'}, \\ a \text{ is a section of } \mathfrak{d} \end{array} \right. \right\}.$$

The two projections

$$(\mathfrak{d}, a) \longmapsto \mathfrak{d} \quad \text{and} \quad (\mathfrak{d}, a) \longmapsto a$$

give morphisms

$$p: X \longrightarrow \mathcal{D} \quad \text{and} \quad \psi: X \longrightarrow W(\mathfrak{g}).$$

When (C_0) holds, let U also denote the open subset of regular points of $W(\mathfrak{g})$ (2.10). Condition (C_2) is then equivalent to saying that the morphism

$$\psi^{-1}(U) \longrightarrow U$$

induced by ψ is an isomorphism; a priori it is a monomorphism by 2.6.

It is immediate that

$$p: X \longrightarrow \mathcal{D}$$

is representable by a projection of vector bundles. More explicitly, for every S -morphism $S' \rightarrow \mathcal{D}$ corresponding to a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}_{S'}$, the fiber product

$$X \times_{\mathcal{D}} S'$$

is represented by the vector bundle $W(\mathfrak{d})$ over S' . Thus, if $\mathcal{D}$ is representable, then so is X . We have:

Theorem 2.16. *Let S be a prescheme and let $\mathfrak{g}$ be a Lie algebra over S whose underlying $\mathcal{O}_S$ -module is locally free of finite type. Suppose that condition (C_0) of 2.9 holds.*

a) *The functor $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$, defined above, is represented by a quasi-projective prescheme of finite presentation over S . The same holds for the functor X .*

b) *If condition (C_2) of 2.9 holds, then $\mathcal{D}$ and X are smooth over S , and the morphism*

$$\psi^{-1}(U) \longrightarrow U$$

induced by ψ is an isomorphism.

c) *Continue to suppose that (C_2) holds. Let $s \in S$, and let $\mathfrak{d}_0$ be a Cartan subalgebra of $\mathfrak{g}(s)$, corresponding to a $k(s)$ -rational point d of $\mathcal{D}$. Suppose that $\mathfrak{d}_0$ contains a regular element of $\mathfrak{g}(s)$, a condition automatically satisfied when $k(s)$ is infinite. If r is the infinitesimal rank of $\mathfrak{g}(s)$ and n its rank over $k(s)$, then there exists an open neighborhood V of d in $\mathcal{D}$ that is S -isomorphic to an open subset V' of*

$$S[t_1, \dots, t_{n-r}].$$

Proof. We may suppose that $\mathfrak{g}$ has constant rank n and constant nilpotent rank r . The assertions concerning X in a) and b) follow immediately from those concerning $\mathcal{D}$ and from the fact that X is a vector bundle over $\mathcal{D}$, defined by a locally free module; they were included only for completeness.

a) The functor $\mathcal{D}$ is a subfunctor of $\text{Grass}_{n-r}(\mathfrak{g})$, whose value at S' is the set of locally free quotient modules of rank $n-r$ of $\mathfrak{g}_{S'}$. It is well known that the latter functor is represented by a projective prescheme smooth over S ; see, for example, Cartan Seminar 1960/61, Exposé 12, Nos. 2 and 3, whose constructions carry over unchanged to preschemes.*⁴

We are therefore reduced to the following relative problem. Given a locally free quotient module of rank $n-r$ of $\mathfrak{g}$, or, equivalently, a locally free submodule $\mathfrak{d}$ of rank r that is locally a direct factor, represent the functor

$$F(S') = \begin{cases} \emptyset, & \text{if } \mathfrak{d}_{S'} \text{ is not a Cartan subalgebra of } \mathfrak{g}_{S'}, \\ \{\emptyset\}, & \text{otherwise.} \end{cases}$$

We shall show that F is represented by a subprescheme of finite presentation over S . This will show that

$$\mathcal{D} \longrightarrow \text{Grass}_{n-r}(\mathfrak{g})$$

*See also EGA I, second edition (announced as forthcoming from North-Holland Publishing Company).

⁴Editors' note: See §I.1.3 of M. Demazure and P. Gabriel, *Groupes algébriques*, Masson (1970).

is represented by an immersion of finite presentation and will complete the proof of a).

First express the condition that $\mathfrak{d}_{S'}$ be a Lie subalgebra of $\mathfrak{g}_{S'}$. This condition says precisely that $S' \rightarrow S$ factors through a certain closed subprescheme S_1 of finite presentation over S ; locally on S , its equations are obtained at once from a basis of $\mathfrak{g}$ adapted to $\mathfrak{d}$. We may therefore suppose that $S = S_1$.

Next we must express that $\mathfrak{d}_{S'}$ contains, fpqc-locally, a quasi-regular section of $\mathfrak{g}_{S'}$. For this purpose consider

$$V = W(\mathfrak{d}) \cap U,$$

where U is the open subset of regular points of $W(\mathfrak{g})$ (2.10). Since the structure morphism $V \rightarrow S$ is smooth and quasi-compact, its image S_1 is an open subset of S , and the immersion $S_1 \rightarrow S$ is quasi-compact, hence of finite presentation. The condition on S' is exactly that $S' \rightarrow S$ factor through S_1 . We are thus reduced to the case $S = S_1$, and, by descent, to the case in which $\mathfrak{d}$ has a section a that is quasi-regular in $\mathfrak{g}$.

It remains to express that, for the induced section $a_{S'}$, the endomorphism

$$\mathrm{ad}(a_{S'})_{\mathfrak{g}_{S'}/\mathfrak{d}_{S'}}$$

is bijective. This again says that $S' \rightarrow S$ factors through an open subprescheme of finite presentation, namely S_D , where

$$D = \det(\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}).$$

But $\mathfrak{d}|_{S_D}$ is then a Cartan subalgebra of $\mathfrak{g}|_{S_D}$. Consequently S_D represents F , which proves a).

b) This follows immediately from 2.11(a) and XI 1.5. It is also, of course, a consequence of the more precise assertion c).

c) Let a_0 be a regular point of $\mathfrak{g}(s)$ contained in $\mathfrak{d}_0$, and extend it to a section a of $\mathfrak{g}$ over an open neighborhood of s . We may plainly suppose that this neighborhood is all of S . Choose a complement M_0 of $\mathfrak{d}_0$ in the vector space $\mathfrak{g}(s)$. After again replacing S by an open neighborhood of s , there is a submodule M of $\mathfrak{g}$, locally a direct factor, such that $M(s) = M_0$; we again suppose that the neighborhood is S .

Let $\mathcal{V}$ be the subfunctor of $\mathcal{D}$ for which $\mathcal{V}(S')$ consists of the Cartan subalgebras $\mathfrak{d}'$ of $\mathfrak{g}_{S'}$ satisfying:

1°) $\mathfrak{d}'$ is a complement of $M_{S'}$;

2°) the unique section in $(a_{S'} + M_{S'}) \cap \mathfrak{d}'$ is a regular section of $\mathfrak{g}_{S'}$.

Condition 1°) defines an open V_1 of $\mathcal{D}$, induced by the open of $\mathrm{Grass}_{n-r}(\mathfrak{g})$ given by the same condition. By 2.10 and (C₂), the conjunction of 1°) and 2°) defines an open of V_1 . Thus $\mathcal{V}$ is represented by an open subprescheme V of $\mathcal{D}$, plainly containing the point d .

Now let $\mathcal{V}'$ be the subfunctor of $W(M)$ defined by

$$\mathcal{V}'(S') = \left\{ u' \in \Gamma(S', M_{S'}) \left| \begin{array}{l} \text{(i) } a_{S'} + u' \text{ is a regular section of } \mathfrak{g}_{S'}, \\ \text{(ii) the unique Cartan subalgebra } \mathfrak{d}' \text{ of } \mathfrak{g}_{S'} \\ \text{containing } a_{S'} + u' \text{ is a complement of } M_{S'} \end{array} \right. \right\}.$$

Condition (i) defines an open subprescheme V'_1 of $W(M)$: it is the inverse image of the open U of regular points of $W(\mathfrak{g})$ (2.10) under the translation morphism $m \mapsto a + m$. The

conjunction of (i) and (ii) defines an open V' of V'_1 , namely the inverse image of V under the evident morphism

$$V'_1 \longrightarrow \mathcal{D}$$

that associates with u' the unique Cartan subalgebra $\mathfrak{d}'$ containing $a_{S'} + u'$. Its restriction gives a morphism

$$V' \longrightarrow V,$$

which is plainly an isomorphism. This proves c).

Corollary 2.17. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field k . Then the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$ (2.16(a)) is quasi-projective, smooth, and irreducible. If $\mathfrak{g}$ contains a regular element (for example, if k is infinite), then $\mathcal{D}$ is a rational variety; that is, its function field is a purely transcendental extension of k .*

Indeed, $\mathcal{D}$ is irreducible because there is a surjective morphism

$$\psi^{-1}(U) \longrightarrow \mathcal{D},$$

and $\psi^{-1}(U)$ is irreducible, being isomorphic to the open subset U of $W(\mathfrak{g})$. The assertion concerning the function field follows immediately from 2.16(c).

Remarks 2.18. I do not know whether this conclusion remains true when k is finite without assuming that $\mathfrak{g}$ contains a regular point; compare 2.12. It can be proved when $\mathfrak{g}$ is the Lie algebra of an algebraic group G smooth over k , at least when $G/\text{radical}(G)$ is an “adjoint” semisimple group, by using a result of Chevalley cited below (see the Appendix). It is plausible that the result remains valid without restriction on G ; for this it would suffice that the cited result of Chevalley be proved for every semisimple algebraic group, not necessarily an adjoint group.

3 Subgroups of type (C) of group preschemes over an arbitrary prescheme

Theorem 3.1. *Let S be a prescheme, let G be a smooth S -group prescheme, and let $\mathfrak{g}$ be its Lie algebra, which is a locally free module of finite type over S . Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ that, as a module, is locally a direct factor in $\mathfrak{g}$, and suppose that, for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Let a be a quasi-regular section of $\mathfrak{g}$ (2.5). Then*

$$M_a = \text{Transp}_G(a, \mathfrak{h}),$$

the subfunctor of G whose S' -valued points are the $g \in G(S')$ such that

$$\text{ad}(g) \cdot a_{S'} \in \Gamma(S', \mathfrak{h}_{S'}),$$

is represented by a closed subprescheme of G , smooth over S , whose structure morphism to S is surjective. ⁵

⁵Editors' note: This transporter is denoted $\text{Transp}_G(a, \mathfrak{h})$.

Proof. Consider the canonical morphism

$$\varphi: G \times_S W(\mathfrak{h}) \longrightarrow W(\mathfrak{g}), \quad (g, x) \longmapsto \operatorname{ad}(g) \cdot x.$$

Then M_a is S -isomorphic to $\varphi^{-1}(a)$, the inverse image under φ of a , viewed as a section of $W(\mathfrak{g})$ over S . To prove that M_a is smooth, it is enough to show that φ is smooth at the points of $G \times_S W(\mathfrak{h})$ lying over $\operatorname{Im}(a)$. More generally, φ is smooth at every point lying over a point of $W(\mathfrak{g})$ that is regular in its fiber $W(\mathfrak{g}(s))$ over S .

Indeed, the source and target of φ are smooth over S , and hence are flat and locally of finite presentation over S . We may therefore check the assertion fiber by fiber. This reduces us to the case in which S is the spectrum of an algebraically closed field k , G is a locally algebraic group over k , $\mathfrak{h}$ is a Lie subalgebra of its Lie algebra $\mathfrak{g}$ containing a Cartan subalgebra of $\mathfrak{g}$, and a is a regular point of $\mathfrak{g}$.

Since φ is a G -morphism, we may suppose that the point of $G \times W(\mathfrak{h})$ under consideration is of the form (e, a) . We may also suppose that G is connected, and hence of finite type over k . The assertion is then precisely XIII 5.4.

It is also immediate that M_a is a closed subscheme of G of finite presentation over S : it is the inverse image of $W(\mathfrak{h})$ under the morphism

$$G \longrightarrow W(\mathfrak{g}), \quad g \longmapsto \operatorname{ad}(g) \cdot a.$$

Finally, the surjectivity of $M_a \rightarrow S$ likewise reduces to the case of an algebraically closed base field. By hypothesis, $\mathfrak{h}$ then contains a Cartan subalgebra $\mathfrak{d}'$. By the conjugacy theorem XIII 6.1(a), $\mathfrak{d}'$ is conjugate to

$$\mathfrak{d} = \operatorname{Nil}(a, \mathfrak{g}).$$

It follows that $\mathfrak{h}$ contains a conjugate of a . This completes the proof.

Corollary 3.2. *Let G and $\mathfrak{g}$ be as in 3.1, with G of finite type over S , and let $\mathfrak{k}$ and $\mathfrak{h}$ be two Lie subalgebras of $\mathfrak{g}$ that are locally direct factors as modules. Suppose that one of the following two hypotheses holds:*

- a) *For every $s \in S$, the geometric fiber $\mathfrak{k}_{\bar{s}}$ is nilpotent and contains a regular element of $\mathfrak{g}_{\bar{s}}$, while the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.*
- b) *$\mathfrak{k}$ is a Cartan subalgebra of $\mathfrak{g}$.*

Then $\operatorname{Transp}_G(\mathfrak{k}, \mathfrak{h})$ is a closed subscheme of G , smooth over S . Moreover, in case a), its structure morphism to S is surjective.

The representability of the transporter by a closed subscheme of G of finite presentation is immediate and is left to the reader. We first prove smoothness in case a). Suppose first that $\mathfrak{k}$ has a section a that is quasi-regular in $\mathfrak{g}$. It is then enough to apply 3.1 and the following lemma.

Lemma 3.3. *Under the hypotheses of 3.2(a), if a is a section of $\mathfrak{k}$ that is quasi-regular in $\mathfrak{g}$, then*

$$\operatorname{Transp}_G(\mathfrak{k}, \mathfrak{h}) = \operatorname{Transp}_G(a, \mathfrak{h}).$$

Indeed, by the definitions it is enough to show that, if a is also a section of $\mathfrak{h}$, then $\mathfrak{k} \subset \mathfrak{h}$. Since $\mathfrak{k}$ is locally nilpotent by hypothesis, 2.1 gives

$$\mathfrak{k} \subset \operatorname{Nil}(a, \mathfrak{g}).$$

On the other hand,

$$\mathrm{Nil}(a, \mathfrak{g}) \subset \mathfrak{h},$$

because

$$\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$$

is injective; this can be checked fiber by fiber by XIII 4.8(b). The lemma follows.

In the general case, the usual standard argument first reduces us to the case where S is affine noetherian, and then to the case where S is local artinian, smoothness being an infinitesimal property. By flat descent we may further suppose that the residue field is infinite. The fiber of $\mathfrak{k}$ then contains an element regular in the corresponding fiber of $\mathfrak{g}$. Lifting this element to a section of $\mathfrak{k}$ reduces us to the preceding case.

Thus the transporter is smooth in case a). Its structure morphism is surjective because this may be checked after reducing to the case where S is the spectrum of an algebraically closed field; $\mathfrak{k}$ then contains a regular point of $\mathfrak{g}$, and 3.3 and 3.1 apply.

For case b), the definition of smoothness (XI 1.1) reduces the assertion to the following lifting statement. Let S be affine, let S_0 be a closed subscheme defined by a quasi-coherent nilpotent ideal J , and let $g_0 \in G(S_0)$ transport $\mathfrak{k}_0$ into $\mathfrak{h}_0$. Then g_0 lifts to an element $g \in G(S)$ that transports $\mathfrak{k}$ into $\mathfrak{h}$. The hypothesis on g_0 places this lifting problem under case a), already proved. This completes the proof.

Of course, if in 3.2(b) the subalgebra $\mathfrak{h}$ satisfies the stronger hypothesis of 3.1, then, and only then, the structure morphism

$$\mathrm{Transp}_G(\mathfrak{k}, \mathfrak{h}) \longrightarrow S$$

is surjective. Using Hensel's lemma XI 1.10, one obtains from 3.1 and 3.2 the following corollary.

Corollary 3.4. *Under the hypotheses of 3.1 for G and $\mathfrak{h}$, and assuming that G is of finite type over S , the following assertions hold:*

- a) *For every quasi-regular section a of $\mathfrak{g}$, étale locally there is a conjugate of a that is a section of $\mathfrak{h}$.*
- b) *For every Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$, étale locally $\mathfrak{d}$ is conjugate to a subalgebra of $\mathfrak{h}$.*

In particular, when $\mathfrak{h}$ is itself a Cartan subalgebra of $\mathfrak{g}$, one obtains the following.

Corollary 3.5. *Let G be a smooth S -prescheme in groups of finite type, let $\mathfrak{g}$ be its Lie algebra, and let $\mathfrak{d}$ and $\mathfrak{d}'$ be two Cartan subalgebras of $\mathfrak{g}$. Then $\mathrm{Transp}_G(\mathfrak{d}, \mathfrak{d}')$ is identical to the strict transporter of $\mathfrak{d}$ into $\mathfrak{d}'$, and is a closed subprescheme of G , smooth over S , whose structure morphism is surjective. Étale locally, $\mathfrak{d}$ and $\mathfrak{d}'$ are conjugate.*

Here the transporter is identical to the strict transporter simply because $\mathfrak{d}$ and $\mathfrak{d}'$ are locally direct factors in $\mathfrak{g}$ and have the same rank at every point. Thus 3.5 is a special case of 3.4. Taking $\mathfrak{d} = \mathfrak{d}'$ gives the following.

Corollary 3.6. *Let G be as in 3.5, and let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Then $\mathrm{Norm}_G(\mathfrak{d})$ is a closed subprescheme in groups of G , smooth over S , and its Lie algebra is $\mathfrak{d}$.*

Indeed, the last assertion says that $\mathfrak{d}$ is its own normalizer in $\mathfrak{g}$, and this follows at once because it is true fiber by fiber.

Corollary 3.7. *Let G and $\mathfrak{g}$ be as in 3.5. Then the conditions (C_2) , (C'_1) , and (C_1) of 2.9 are equivalent. In other words, if $\mathfrak{g}$ admits a Cartan subalgebra fpqc locally, then every quasi-regular section of $\mathfrak{g}$ is regular.*

Indeed, let a be a quasi-regular section; we prove that it is regular. Since the question is fpqc local, we may suppose that $\mathfrak{g}$ admits a Cartan subalgebra $\mathfrak{d}$. By 3.4(a), étale locally a is conjugate to a section of $\mathfrak{d}$. We are thereby reduced to the case in which a is a section of $\mathfrak{d}$, where the conclusion follows immediately from the definition.

Definition 3.8. *Let G be a smooth prescheme in groups over a prescheme S . A subgroup of type (C) of G is a subprescheme in groups $D \subset G$, smooth over S and with connected fibers, such that*

$$\mathfrak{d} = \text{Lie}(D)$$

is a Cartan subalgebra (2.4) of $\mathfrak{g} = \text{Lie}(G)$. Equivalently, for every $s \in S$, the fiber D_s is a subgroup of type (C) of the algebraic group G_s (XIII 6.2).

Theorem 3.9. *Let G be a smooth S -prescheme in groups of finite presentation, and let $\mathfrak{g}$ be its Lie algebra. Then:*

a) *The map*

$$D \longmapsto \mathfrak{d} = \text{Lie}(D)$$

establishes a bijective correspondence between the subgroups of type (C) of G and the Cartan subalgebras of $\mathfrak{g}$.

b) *If D and $\mathfrak{d}$ correspond, then*

$$\text{Norm}_G(D) = \text{Norm}_G(\mathfrak{d}).$$

This is a closed subprescheme of G , smooth over S , and

$$D = \text{Norm}_G(D)^0 = \text{Norm}_G(\mathfrak{d})^0.$$

c) *Any two subgroups D and D' of type (C) in G are conjugate étale locally.*

Proof. Let D be a subgroup of type (C) of G , and let $\mathfrak{d}$ be its Lie algebra. Then

$$D \subset \text{Norm}_G(\mathfrak{d}).$$

By Definition 3.8 and Corollary 3.6, this is an inclusion of smooth preschemes in groups over S that induces an isomorphism on Lie algebras. Since D has connected fibers, it follows that

$$D = \text{Norm}_G(\mathfrak{d})^0.$$

Thus the map in a) is injective. We prove that it is surjective. Let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. By 3.6,

$$N = \text{Norm}_G(\mathfrak{d})$$

is a closed subprescheme in groups of G , smooth over S , and has Lie algebra $\mathfrak{d}$. Since G is of finite presentation over S , so is N . Consequently, as noted in XII after 7.3, the union of the connected components of the fibers of N is the underlying set of an open subprescheme in groups $N^\circ \subset N$. Evidently N° is a subgroup of type (C) of G with Lie algebra $\mathfrak{d}$. This proves a). The first assertion of b) follows at once, and the formula

$$D = \text{Norm}_G(\mathfrak{d})^0$$

has already been proved. Finally, c) follows from a) and 3.5.

Corollary 3.10. *Suppose that $\mathfrak{g}$ admits a Cartan subalgebra fpqc locally, or equivalently by 3.9(a), that G admits a subgroup of type (C) fpqc locally. Consider the functor*

$$\mathcal{D}: (\text{Sch})^\circ/S \longrightarrow (\text{Sets}),$$

where $\mathcal{D}(S')$ is the set of subgroups of type (C) of $G_{S'}$. This functor is representable by a quasi-projective prescheme, smooth over S , with connected geometric fibers. If $S = \text{Spec}(k)$, so that G is a smooth algebraic group over k , and if $\mathfrak{g}$ admits a regular point—a condition automatically satisfied when k is infinite—then $\mathcal{D}$ is a rational variety over k .

Indeed, by 3.9(a) the functor $\mathcal{D}$ is canonically isomorphic to the functor considered in 2.16. On the other hand, by 3.7 condition (C₂) is satisfied. Therefore 3.10 follows from 2.16 and 2.17.

Corollary 3.11. *Let D be a subgroup of type (C) of G , and let N be its normalizer in G . Then the quotient sheaf G/N is canonically isomorphic to the functor $\mathcal{D}$ of 3.10. It is therefore representable by a quasi-projective prescheme, smooth over S , with connected geometric fibers.*

Proposition 3.12. *Let G be a smooth S -prescheme in groups of finite presentation over S . Let H and K be two subpreschemes in groups, smooth and of finite presentation over S , with K having connected fibers, and let $\mathfrak{g}, \mathfrak{h}, \mathfrak{k}$ be the corresponding Lie algebras. Suppose that one of the following conditions holds on their geometric fibers:*

- a) *For every $s \in S$, $\mathfrak{k}_{\bar{s}}$ contains a regular element of $\mathfrak{g}_{\bar{s}}$, and $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.*
- b) *For every $s \in S$, $\mathfrak{k}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.*

Under these conditions, $H \supset K$ if and only if $\mathfrak{h} \supset \mathfrak{k}$.

It remains only to prove that $\mathfrak{h} \supset \mathfrak{k}$ implies $H \supset K$. In case b), the inclusion $\mathfrak{h} \supset \mathfrak{k}$ places us under the hypotheses of case a), so it is enough to prove a). Proceeding as in 3.2(a), by reduction to the case where S is local artinian, we reduce to the case in which there exists a section a of $\mathfrak{k}$ that is quasi-regular in $\mathfrak{g}$. Arguing then as in XIII 5.5 reduces us to the following statement.

Corollary 3.13. *Let G be a smooth S -prescheme in groups of finite presentation over S , let $H \subset G$ be a subprescheme in groups smooth and of finite presentation over S , and let $\mathfrak{g}$ and $\mathfrak{h}$ be their Lie algebras. Let a be a section of $\mathfrak{h}$ that is quasi-regular in $\mathfrak{g}$. Suppose that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Put*

$$M_a = \text{Transp}_G(a, \mathfrak{h}).$$

This is a closed subprescheme of G , smooth over S (cf. 3.1). Hence M_a^0 , the union in the fibers of M_a of the connected components containing the identity, is an open subset of M_a , endowed with the structure induced from M_a . Then

$$H^0 = M_a^0.$$

Clearly $H \subset M_a$, and therefore $H^0 \subset M_a^0$. Since this is an inclusion of preschemes smooth over S , it is enough to check equality on the fibers. This reduces us to the case where S is the spectrum of an algebraically closed field, which was treated in XIII 5.4.

Corollary 3.14. *Let G be a smooth S -prescheme in groups of finite presentation, let D be a subgroup of type (C) of G , and let $H \subset G$ be a subprescheme in groups smooth and of finite presentation over S . If $\mathfrak{d}$ and $\mathfrak{h}$ are their Lie algebras, then*

$$\mathrm{Transp}_G(D, H) = \mathrm{Transp}_G(\mathfrak{d}, \mathfrak{h}),$$

and this functor is representable by a closed subprescheme of G , smooth over S .

Indeed, the identity between the two transporters follows from 3.12(b), after which 3.2 applies.

Corollary 3.15. *Let G and H be as in 3.14, and suppose that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Suppose also that $\mathfrak{g}$ admits a Cartan subalgebra fpqc locally. Then, étale locally, H contains a subgroup of type (C) of G .*

By 3.7 and 3.9(a), G admits a subgroup of type (C) étale locally. We may therefore suppose that G admits such a subgroup D . The hypothesis on $\mathfrak{h}$ then also says that the structure morphism of the transporter in 3.14 is surjective, by the conjugacy theorem XIII 6.1(a). The conclusion follows from Hensel's lemma XI 1.10.

Corollary 3.16. *Let G, H, K be as in 3.12(a), and suppose in addition that every geometric fiber $\mathfrak{k}_{\bar{s}}$ is nilpotent, that is, that $\mathfrak{k}$ is locally nilpotent. Then*

$$\mathrm{Transp}_G(K, H) = \mathrm{Transp}_G(\mathfrak{k}, \mathfrak{h}),$$

and this functor is representable by a closed subprescheme of G , smooth over S , whose structure morphism to S is surjective. Étale locally, H contains a subgroup conjugate to K .

The identity between the two transporters is again contained in 3.12(a). The assertion about its structure is precisely 3.2(a), and the last assertion of 3.16 then follows from Hensel's lemma.

Remarks 3.17. a) In 3.12 and 3.16, the hypothesis that K be smooth over S may be replaced by the weaker hypothesis that the sheaf

$$e_K^*(\Omega_{K/S}^1) = \omega_K^1$$

of relative 1-differentials along the identity section be locally free. In this form, 3.12 includes XIII 5.5.

b) Let $G, \mathfrak{g}, \mathfrak{h}$ be as in 3.1, with G of finite presentation over S . Then

$$N = \mathrm{Norm}_G(\mathfrak{h})$$

need not be smooth over S along the identity section. Equivalently, there need not exist a subprescheme in groups $H \subset G$, smooth over S , whose Lie algebra is $\mathfrak{h}$, even when S is the spectrum of a field. When such an H exists, taking H with connected fibers gives

$$N = \mathrm{Norm}_G(H);$$

I do not know whether N is then smooth over S . In this question one may plainly reduce to the case where S is local artinian.

To conclude this section, we examine the case in which G is “semisimple” over S .

Theorem 3.18. *Let S be a prescheme and let G be a smooth S -prescheme in groups whose geometric fibers are semisimple “adjoint” groups, that is, semisimple groups whose reductive center (XII 4.1 and 4.4) is the identity group. Then the subgroups of type (C) of G are precisely its maximal tori, and hence also precisely its Cartan subgroups (XII 3.1).*

In view of the definitions, it is enough to treat the case where S is the spectrum of an algebraically closed field. We must then show that, for a maximal torus T of G , its Lie algebra $\mathfrak{t}$ is a Cartan subalgebra of $\mathfrak{g}$. Put $r = \dim T = \text{rank } \mathfrak{t}$. Taking into account the inequality

$$\text{nil rank}(\mathfrak{g}) \geq \text{nil rank}(G) = \dim T = \text{rank } \mathfrak{t} = r,$$

it is enough to find $x \in \mathfrak{t}$ such that

$$\dim \text{Nil}(x, \mathfrak{g}) = r.$$

Since $\mathfrak{t}$ is abelian, and therefore nilpotent, this is equivalent to finding $a \in \mathfrak{t}$ for which

$$\text{ad}(a)_{\mathfrak{g}/\mathfrak{t}}$$

is injective (XIII 5.7(a)).

Consider the characters α of T occurring in the representation of T induced by the adjoint representation of G . The structure theory of the semisimple group G (*Bible*), 13, Theorem 1(a), and Theorem 3, Corollary 2, or more precisely the “big cell” of G , shows that the eigenspace of $\mathfrak{g}$ for the trivial character is exactly $\mathfrak{t}$. The other eigenspaces are one-dimensional, and their characters are precisely the roots of G relative to T . Here the big cell is built from T and the subgroups P_α , each isomorphic to the additive group $\mathbf{G}_a$, stable under T , and corresponding to the root characters.

The reductive center of G is the intersection of the kernels of the characters of T occurring in the adjoint representation (XII 4.8). Thus the assertion that G is adjoint says that its roots generate the lattice

$$M = \text{Hom}(T, \mathbf{G}_m).$$

A familiar lemma from root theory says that every root belongs to a system of simple roots, hence to a basis of the group generated by the roots, and therefore to a basis of the dual lattice of T .⁶ We obtain the following.

Corollary 3.19. *Let G be an adjoint semisimple algebraic group over an algebraically closed field k , and let T be a maximal torus of G . For every root*

$$\alpha: T \longrightarrow \mathbf{G}_m$$

of G relative to T , the corresponding homomorphism

$$\alpha': \mathfrak{t} \longrightarrow k$$

is nonzero.

This assertion is essentially equivalent to Theorem 3.18. Indeed, for $t \in \mathfrak{t}$, the endomorphism $\text{ad}(t)$ is semisimple, and its eigenvalues on $\mathfrak{g}/\mathfrak{t}$ are precisely the numbers $\alpha'(t)$. Hence $\text{ad}(t)_{\mathfrak{g}/\mathfrak{t}}$ is injective if and only if all the $\alpha'(t)$ are nonzero; and such a $t \in \mathfrak{t}$ exists if and only if every α' is nonzero.

⁶The re-edition replaces $\widehat{T}$ by T at this point.

Corollary 3.20. *Let S be a prescheme and let G be a smooth S -prescheme in groups of finite presentation over S , whose geometric fibers are connected reductive algebraic groups, that is, extensions of a semisimple group by a torus. For every $s \in S$ there is an open neighborhood U of s such that $G|_U$ admits a maximal torus.⁷*

We shall see in Exposé XVI that the hypothesis on G implies that G is affine over S and has locally constant reductive rank. Hence, by XII 4.7(c), G admits a reductive center Z , the quotient

$$G' = G/Z$$

is a smooth affine group over S whose reductive center is the identity group, and the maximal tori of G correspond bijectively to those of G' . Moreover, the geometric fibers of G' are plainly connected semisimple groups, and are adjoint by definition because their reductive center is trivial.

It is therefore enough to treat the case where G is semisimple and adjoint. By 3.18, the problem reduces to finding an open neighborhood U of s and a subgroup of type (C) of $G|_U$, or equivalently by 3.9(a), a Cartan subalgebra of $\mathfrak{g}|_U$. This is possible when $k(s)$ is infinite: by 3.7, $\mathfrak{g}$ satisfies condition (C₂) of 2.9, and 2.11(b) applies.

In fact, 3.20 remains true without assuming that $k(s)$ is infinite. By the preceding argument, it is enough to know that, for every adjoint semisimple group G over a finite field k , the Lie algebra $\mathfrak{g}$ contains a regular element. Chevalley proved this assertion using the properties of a Coxeter element of the Weyl group; see the appendix below by J.-P. Serre.

Remarks 3.21. a) Assertion 3.20 remains valid, with essentially the same proof, if G is replaced by a closed subprescheme in groups H , smooth over S , whose rank is everywhere the same as that of G (for example, a “parabolic subgroup” of G), provided that $k(s)$ is infinite. I do not know whether the infinitude hypothesis is again superfluous. It can at least be removed when H is parabolic: constructing the radical U of H and the quotient H/U reduces the question to the semisimple case. Unfortunately, the method of regular elements seems powerless here. One readily constructs examples—for instance, the projective group with its “standard” Borel subgroup—in which the Lie algebra $\mathfrak{h}$ of H contains no regular element.

b) When $k(s)$ is infinite, the proof of 3.20 in fact gives the following sharper statement, using 3.9(b): every maximal torus T_0 of G_s is induced by a maximal torus T over an open neighborhood of s . I do not know whether this remains true without the infinitude hypothesis. The evident difficulty is that the Lie algebra $\mathfrak{t}_0$ of T_0 need not contain a regular element of the Lie algebra $\mathfrak{g}_0$ of the fiber G_s .

An affirmative answer would imply the following assertion, which is known only when the residue field is infinite or when A is separated and complete. Let A be a local ring with residue field k , and let M be an “Azumaya algebra” over A , meaning an algebra that is a finite free A -module and whose reduction

$$M_0 = M \otimes_A k$$

is a central simple algebra over k . Let $D_0 \subset M_0$ be a commutative subalgebra separable over k , with

$$[M_0 : k] = [D_0 : k]^2.$$

⁷This result, as well as 2.11 on which it rests, immediately generalizes to the case in which s is replaced by a finite subset of S contained in an affine open.

Then there should exist a commutative subalgebra $D \subset M$ that is a direct summand as an A -module and satisfies

$$D \otimes_A k = D_0.$$

The datum of M is equivalent to that of a principal homogeneous bundle under the projective group $\mathrm{PGL}(n)_A$, and hence to an inner twisted form G of $\mathrm{PGL}(n)$. The maximal tori of G correspond bijectively to the commutative subalgebras $D \subset M$ that are étale of rank n over A .

c) Applying 3.20 to the centralizer of a subtorus Q of a reductive S -prescheme in groups G , one sees that every such Q is contained, Zariski locally, in a maximal torus of G .

4 A Digression on Borel Subgroups

Definition 4.1. *Let G be a smooth algebraic group over an algebraically closed field. A Borel subgroup of G is a smooth, solvable, connected algebraic subgroup that is maximal with these properties.*

When G is affine, this agrees with the terminology of (Bible), 6, Definition 1. Observe at once that if Z is a connected smooth subgroup of G contained in its center (or, more generally, if Z is solvable and normal), then, for every Borel subgroup B of G , the image BZ of $B \times Z$ under the morphism

$$(b, z) \longmapsto bz$$

from $B \times Z$ to G is a smooth, solvable, connected subgroup of G containing B . It is therefore equal to B . Thus B contains Z , and consequently B is the inverse image of an algebraic subgroup B' of $G' = G/Z$; it is immediate that B' is a Borel subgroup of G' .

Taking

$$Z = \mathrm{Centr}(G^0)_{\mathrm{red}}^0,$$

the quotient $G' = G/Z$ is affine by XII 6.1. Hence the Borel subgroups of G' are conjugate, and G'/B' is a projective variety for every such B' (Bible), 6, Theorem 4. Consequently:

Proposition 4.2. *Let G be as in 4.1. Then the Borel subgroups of G are conjugate. If B is a Borel subgroup, then G/B is a projective variety. The maximal tori of B are maximal tori of G ; and, when G is connected, the Cartan subgroups of B are Cartan subgroups of G .*

It remains to prove the last assertion, and we may plainly suppose that $G = G^0$. For Cartan subgroups, it follows from the analogous assertion in G' (Bible), 6, Theorem 4, Corollary 4 and XII 6.6(e). The assertion for maximal tori follows from this one, since by XII 6.6(c) the maximal tori of a smooth algebraic group are precisely the maximal tori of its Cartan subgroups.

Corollary 4.3. *Suppose that G is connected. Then every element of G is contained in a Borel subgroup of G .*

This reduces to the same assertion in G' , where it is well known (Bible), 6, Theorem 5(d).

Corollary 4.4. *Let B be a Borel subgroup of G , let C be a Cartan subgroup of B , and let N be its normalizer in G . Then*

$$N \cap B = C.$$

Indeed,

$$N \cap B = \text{Norm}_B(C),$$

so it remains to show that, when G is connected and solvable, a Cartan subgroup C is its own connected normalizer. With the preceding notation, C is the inverse image of a Cartan subgroup C' of G' , and we are therefore reduced to the case in which G is affine. The normalizer of a Cartan subgroup is smooth (C being its own connected normalizer; see, for example, XII 6.6(c)). It is thus enough to check that C and N have the same k -valued points, which is exactly (*Bible*), Theorem 6(d).

Definition 4.5. *Let G be a smooth prescheme in groups of finite presentation over a prescheme S . A Borel subgroup of G is a smooth subprescheme in groups B of G , of finite presentation, such that for every $s \in S$ the geometric fiber $B_{\bar{s}}$ is a Borel subgroup of $G_{\bar{s}}$.*

As is immediately verified, this notion is stable under base change and is local for the fpqc topology. Indeed, if k' is an algebraically closed extension of an algebraically closed field k , then an algebraic subgroup B of a smooth algebraic group G over k is a Borel subgroup of G if and only if $B_{k'}$ is a Borel subgroup of $G_{k'}$.

It follows from the definition that, if G is a smooth algebraic group over an arbitrary field k and B is a Borel subgroup of G , then G/B is a projective variety; every maximal torus T of B is a maximal torus of G ; and the normalizer of T in B equals its centralizer C , which is a Cartan subgroup of G when G is connected.

Remarks 4.6. Unfortunately, it is no longer true in general that two Borel subgroups of G are locally conjugate for the fpqc topology. This can fail even when G is affine over S and S is the spectrum of the algebra of dual numbers over an algebraically closed field k .

As a consequence of this regrettable fact, let us point out that if G is a smooth, affine, connected algebraic group over a nonperfect field k , it is in general impossible to define naturally a homogeneous space D under G that plays the role of a flag variety, that is, of the variety of Borel subgroups of G . Over the algebraic closure $\bar{k}$ of k , such a space would have to be isomorphic to $G_{\bar{k}}/B$, where B is a Borel subgroup of $G_{\bar{k}}$.

Indeed, suppose that the quotient of $G_{\bar{k}}$ by its radical $\bar{R}$ is an adjoint semisimple group. The kernel of

$$G_{\bar{k}} \longrightarrow \text{Aut}_{\bar{k}}(\bar{D})$$

is then the radical of $G_{\bar{k}}$. Consequently, if $\bar{D}$ came from a homogeneous space D under G , the radical $\bar{R}$ would come from a subgroup R of G . But examples are easily constructed for which $G_{\bar{k}}/\bar{R}$ is adjoint while $\bar{R}$ is not defined over k .

Under these conditions, the functor

$$\mathfrak{B}: (\text{Sch}/S)^{\circ} \longrightarrow \text{Ens}, \quad \mathfrak{B}(S') = \{\text{Borel subgroups of } G_{S'}\},$$

is not representable by a smooth S -prescheme. From the infinitesimal point of view (III, §3), the failure of the conjugacy theorem is expressed by the fact that, if B is a Borel subgroup of the smooth algebraic group G , the cohomology group

$$H^1(B, \mathfrak{g}/\mathfrak{b})$$

may be nonzero.⁸

We shall see, on the other hand, in a later exposé that when G is semisimple, or more generally reductive, such unpleasant phenomena do not occur. These phenomena, together no doubt with the absence of good existence theorems, explain why Borel subgroups play only a relatively unobtrusive role in the scheme-theoretic study of general group schemes, whereas they dominate the theory of semisimple group schemes in the later exposés.

Proposition 4.8. ⁹ *Let G be a smooth S -prescheme in groups of finite presentation with connected fibers, and let B be a Borel subgroup of G . Then B is its own normalizer and is a closed subscheme of G .*

By XII 7.10, it is enough to prove that, over an algebraically closed field k , every element of $G(k)$ that normalizes B belongs to $B(k)$. When G is affine this is a fundamental result of Chevalley (*Bible*), 9, Theorem 1; the general case reduces immediately to it by the reduction already used in 4.2.

Remarks 4.8.1. Definition 4.5 may be generalized by also introducing the notion of a *parabolic subgroup* of G . This means a subscheme in groups P of G , smooth and of finite presentation over S , such that for every $s \in S$ the geometric fiber P_s is a parabolic subgroup of G_s , that is, contains a Borel subgroup of G_s .

Proposition 4.8 extends to a parabolic subgroup P of G , by the same reduction to the known set-theoretic assertion. We note the following consequence (compare Exposé XVI). If P is a parabolic subgroup of G , then G/P is representable by a quasi-projective prescheme of finite presentation over S ; here G is assumed to have connected fibers. Moreover, G/P is plainly smooth over S , and its geometric fibers are connected and proper. It follows readily from EGA III, 5.5.1 that

$$D = G/P$$

is proper, hence projective, over S . If its relative dimension is n , it is known that the inverse of the invertible sheaf $\Omega_{D/S}^n$ induces ample sheaves on the geometric fibers of D/S ; hence, by EGA III, 4.7.1,

$$(\Omega_{D/S}^n)^{-1}$$

is ample on D relative to S .

It is readily seen, by reduction to the affine case and to an algebraically closed base field, that if

$$u: G \longrightarrow G'$$

is an epimorphism of smooth algebraic groups, then $u(B) = B'$ is a Borel subgroup of G' for every Borel subgroup B of G . We now consider the case in which this gives a bijective correspondence between the Borel subgroups of G and those of G' .

Proposition 4.9. *Let G and G' be smooth S -preschemes in groups of finite presentation with connected fibers, and let*

$$u: G \longrightarrow G'$$

be a faithfully flat, that is, surjective, homomorphism of groups.¹⁰ Put $N = \ker u$, and suppose that one of the following two conditions holds:

⁸Here $\mathfrak{b}$ denotes the Lie algebra of B .

⁹There is no item numbered 4.7 in the source.

¹⁰The editors ask whether a reference should be added here.

a) N is central in G .

b) $S = \operatorname{Spec}(k)$ for a field k , and, if $\bar{k}$ is an algebraic closure of k , then

$$N_{\bar{k}} = \ker(u_{\bar{k}})$$

is contained in the radical of $G_{\bar{k}}$, that is, in its largest smooth, connected, solvable, normal subgroup.

Then the map

$$B' \longmapsto u^{-1}(B')$$

induces a bijection from the set of Borel subgroups of G' onto the set of Borel subgroups of G .

Case b) follows immediately from the correspondence between algebraic subgroups of G' and algebraic subgroups of G containing N , together with the fact that, over an algebraically closed field, the Borel subgroups of G contain its radical. The latter fact follows at once from the argument preceding 4.2.

We prove case a). By XII 7.12, the map

$$H' \longmapsto H = u^{-1}(H')$$

gives a bijection between the subschemes in groups H' of G' that are smooth and of finite presentation over S , have connected fibers, and at every $s \in S$ have the same reductive rank and the same nilpotent rank as G' , and the subschemes in groups H of G having the analogous properties. Borel subgroups of G' and of G have these properties.

It remains to show that, when H' and H correspond, H' is a Borel subgroup of G' if and only if H is a Borel subgroup of G . By definition, this question reduces to the case where S is the spectrum of an algebraically closed field. Since N is central in G , and hence in H , the group H is solvable if and only if H' is solvable. Finally, the correspondence between algebraic subgroups of G' and algebraic subgroups of G containing N shows immediately that H' has the maximality property of Definition 4.1 if and only if H does. This completes the proof.

Corollary 4.10. *With the notation of 4.9,¹¹ if B' and B are corresponding Borel subgroups of G' and G , then*

$$\mathfrak{b} = \operatorname{Lie}(u)^{-1}(\mathfrak{b}'),$$

where $\mathfrak{g}, \mathfrak{g}', \mathfrak{b}, \mathfrak{b}'$ are the Lie algebras of G, G', B, B' , respectively, and

$$\operatorname{Lie}(u): \mathfrak{g} \longrightarrow \mathfrak{g}'$$

is the homomorphism induced by u .

This follows immediately from the definitions and the identity

$$u^{-1}(B') = B.$$

We can now prove the principal result of this section.

¹¹The source prints “4.7”; the intended reference is 4.9.

Theorem 4.11. *Let G be a smooth algebraic group over an algebraically closed field k , and let $\mathfrak{g}$ be its Lie algebra. Then $\mathfrak{g}$ is the union of the Lie algebras $\mathfrak{b}$ of the Borel subgroups B of G .*

We may plainly suppose that G is connected. Let R be the radical of G , and put $G' = G/R$. By 4.9(b) and 4.10, it is enough to prove the theorem for G' ; thus we may suppose that G is semisimple.

Let Z be the center of G , which is its reductive center, and put $G' = G/Z$. The same argument, now using 4.9(a), reduces the theorem to G' . We may therefore suppose that G is semisimple and adjoint.

Let B be a Borel subgroup of G , let T be a maximal torus of B , hence of G , and let $\mathfrak{b}$ and $\mathfrak{t}$ be their Lie algebras. By 3.18, T is a subgroup of type (C) of G , that is, $\mathfrak{t}$ is a Cartan subalgebra of $\mathfrak{g}$. The union of the conjugates of $\mathfrak{t}$ is consequently dense in $\mathfrak{g}$ (XIII 5.1, (i) $\Rightarrow$ (vii)). *A fortiori*, the union of the conjugates of $\mathfrak{b}$ is dense in $\mathfrak{g}$.

Let X be the closed subscheme of

$$G/B \times W(\mathfrak{g})$$

whose k -valued points are the pairs (gB, x) such that

$$x \in \text{Ad}(g)\mathfrak{b}$$

(compare XIII 1). The morphism

$$\psi: X \longrightarrow W(\mathfrak{g})$$

induced by the second projection is proper because G/B is proper over k . We have just seen that it is dominant; it is therefore surjective. This proves 4.11.

The only result of this section that we shall use in the remainder of this exposé is the following corollary.

Corollary 4.12. *Let k be an infinite field, let G be a smooth algebraic group over k , and let T be a maximal torus of G . Write $\mathfrak{g}$ and $\mathfrak{t}$ for their Lie algebras. Let*

$$u: G \longrightarrow \text{GL}(V)$$

be a linear representation, where V is finite-dimensional over k , and let

$$u': \mathfrak{g} \longrightarrow \mathfrak{gl}(V)$$

be the induced representation of Lie algebras. Then the minimum of the nullity of $u'(x)$, for $x \in \mathfrak{g}$, is attained at an element $x \in \mathfrak{t}$.

We reduce immediately to the case where k is algebraically closed. We may plainly suppose that G is connected, and, after dividing G by $(\ker u)_{\text{red}}$, that G is affine. Using 4.11 and the conjugacy theorem for the maximal tori of G , we reduce to the further case in which G is solvable.

Then G is a semidirect product $T \cdot U$, where U is the unipotent part of G , a smooth connected unipotent group (*Bible*), 6, Theorem 3. Thus, as a vector space,

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{u}, \quad \mathfrak{u} = \text{Lie}(U).$$

By the Lie–Kolchin theorem (*Bible*), 6, Theorem 1, the representation space V has a composition series by stable subspaces V_i , with one-dimensional quotients

$$W_i = V_i/V_{i+1}.$$

For every i there is therefore an induced representation

$$u_i: G \longrightarrow \mathrm{GL}(W_i) = \mathbb{G}_m$$

and a corresponding homomorphism of Lie algebras

$$u'_i: \mathfrak{g} \longrightarrow k.$$

For every $x \in \mathfrak{g}$, the nullity of $u'_i(x)$ is the number of indices i for which $u'_i(x) = 0$.

Since U is unipotent, every u_i is trivial on U , and hence every u'_i vanishes on $\mathfrak{u}$. If

$$x = t + v, \quad t \in \mathfrak{t}, \quad v \in \mathfrak{u},$$

then $u'_i(x) = u'_i(t)$ for every i . Thus $u'_i(x)$ and $u'_i(t)$ have the same nullity, and 4.12 follows.

5 Relations Between Cartan Subgroups and Cartan Subalgebras

Applying 4.12 to the adjoint representation of G , we obtain:

Theorem 5.1. *Let G be a smooth algebraic group over an infinite field, let T be a maximal torus of G , and let $\mathfrak{g} \supset \mathfrak{t}$ be their Lie algebras. Then $\mathfrak{t}$ contains a regular element of $\mathfrak{g}$.*

Corollary 5.2. *Let G be a smooth S -prescheme in groups of finite presentation, and let $\mathfrak{g}$ be its Lie algebra.*

a) *Conditions (C_0) through (C_2) of 2.9 for $\mathfrak{g}$ are equivalent. In particular, if the infinitesimal rank of the fibers of $\mathfrak{g}$ at the points of S is locally constant, then, étale locally, $\mathfrak{g}$ admits a Cartan subalgebra and hence, by 3.9(a), G admits a subgroup of type (C).*

b) *Let H be a subprescheme in groups of G , smooth and of finite presentation over S , whose fibers are connected and at every $s \in S$ have the same reductive rank as G ; for example, H may be a maximal torus or a Cartan subgroup of G . Let D be a subgroup of type (C) of G . Then*

$$H \subset D \iff \mathfrak{h} \subset \mathfrak{d}.$$

c) *Suppose that condition (C_0) holds, that is, the infinitesimal rank of G is locally constant. Let H be a subprescheme in groups of G , smooth and of finite presentation over S , with connected nilpotent fibers, and having at every point the same reductive rank as G ; for example, H may be a maximal torus or a Cartan subgroup of G . Then H is contained in a unique subgroup D of type (C) of G .*

d) *If G admits a Cartan subgroup fpqc locally, then so does every subgroup D of type (C) of G .*

Proof. a) Assume (C_0) , and let us prove (C_2) , that is, that every quasi-regular section a of $\mathfrak{g}$ is regular. As usual, we reduce first to S affine noetherian and then, since the question is infinitesimal by 2.15(b), to S local Artinian. In this last case (C_0) is automatic. We may moreover suppose that the residue field k of S is infinite.

By 3.7, it is enough to show that $\mathfrak{g}$ admits a Cartan subalgebra. Let T be a maximal torus of G . By 5.1 there is a quasi-regular element t in the Lie algebra $\mathfrak{t}$ of T . We shall show that t is regular.

Consider the linear representation of T on $\mathfrak{g}$ induced by the adjoint representation of G . There is a finite set of characters $(u_i)_{i \in I}$ of T and a direct-sum decomposition

$$\mathfrak{g} = \bigoplus_{i \in I} \mathfrak{g}_i$$

into T -stable submodules such that T acts on $\mathfrak{g}_i$ through u_i (I, §4.7.3). Let

$$u'_i: \mathfrak{t} \longrightarrow A$$

be the homomorphism of Lie algebras induced by $u_i: T \rightarrow \mathbb{G}_m$, where A is the ring of S . Let u_{i0} and u'_{i0} be their reductions under $A \rightarrow k$. Put

$$I' = \{i \in I \mid u'_{i0} \neq 0\}, \quad I'' = I \setminus I'.$$

The regularity of t_0 is expressed by

$$u'_{i0}(t_0) \neq 0 \quad (i \in I'),$$

and therefore $u'_i(t)$ is invertible for $i \in I'$.

The Cartan subalgebra of $\mathfrak{g}_0$ defined by t_0 , that is, the kernel of the semisimple endomorphism $\text{ad}(t_0)$, is

$$\bigoplus_{i \in I''} (\mathfrak{g}_i)_0.$$

Set

$$\mathfrak{d} = \bigoplus_{i \in I''} \mathfrak{g}_i.$$

Then $\text{ad}(t)$ is nilpotent on $\mathfrak{d}$, and is an automorphism of

$$\mathfrak{g}/\mathfrak{d} \simeq \bigoplus_{i \in I'} \mathfrak{g}_i.$$

By 2.6, t is regular.

b) This follows from 3.12(a). By 5.1, H satisfies the required hypothesis that every geometric fiber of $\mathfrak{h}$ contains a regular element of the corresponding fiber of $\mathfrak{g}$.

c) By b), it is enough to show that $\mathfrak{h}$ is contained in a unique Cartan subalgebra of $\mathfrak{g}$. By a), $\mathfrak{g}$ satisfies (C_2) . We are therefore reduced to the following lemma.

Lemma 5.3. *Let $\mathfrak{g}$ be a Lie algebra over S , locally free of finite type as a module, and satisfying condition (C_2) of 2.9. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ such that $\mathfrak{h}$ is locally a direct summand, is locally nilpotent, and, for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a regular element of $\mathfrak{g}_{\bar{s}}$. Then $\mathfrak{h}$ is contained in a unique Cartan subalgebra of $\mathfrak{g}$.*

In the situation at hand, $\mathfrak{h}$ has these properties: it is locally nilpotent because H has nilpotent fibers, and the condition on regular elements follows from 5.1.

Since 5.3 is fpqc local, it is enough, at a point $s \in S$ with $\kappa(s)$ infinite, to find an open neighborhood U of s on which existence and uniqueness hold after every base change $S' \rightarrow S$ factoring through U . Choose a regular element of $\mathfrak{g} \otimes \kappa(s)$ lying in $\mathfrak{h} \otimes \kappa(s)$, and extend it to a section a of $\mathfrak{h}$ over an open neighborhood U of s . Condition (C_2) and 2.10 allow us to shrink U so that a is regular. We may thus suppose $U = S$.

Every Cartan subalgebra of $\mathfrak{g}$ containing $\mathfrak{h}$ contains a , and hence equals

$$\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$$

by 2.6; this proves uniqueness. Conversely, $\mathfrak{h} \subset \mathfrak{d}$ because $\mathfrak{h}$ is locally nilpotent, which proves existence.

d) This is a special case of XII 7.9(d).

Corollary 5.4. *Let G be a smooth connected algebraic group over an algebraically closed field k . Let H be a connected algebraic subgroup such that $\mathfrak{h}$ contains a Cartan subalgebra of $\mathfrak{g}$, or equivalently such that H contains a subgroup of type (C) of G . Then the number of conjugates of H containing a regular element of $G(k)$ equals the number of conjugates of $\mathfrak{h}$ containing a regular element of $\mathfrak{g}$.*

Indeed, let g be a regular element of $G(k)$, and let C be the unique Cartan subgroup of G containing g (XIII 2). The conjugates of H that contain g are exactly those that contain C (XIII 2.8(b)).

Similarly, let x be a regular element of $\mathfrak{g}$. If a conjugate $\mathfrak{h}'$ of $\mathfrak{h}$ contains x , then

$$\mathrm{ad}(x)_{\mathfrak{g}/\mathfrak{h}'}$$

is injective by XIII 5.4, and therefore

$$\mathfrak{h}' \supset \mathrm{Nil}(x, \mathfrak{g}) = \mathfrak{d}.$$

Thus the number of conjugates of $\mathfrak{h}$ containing x equals the number containing the Cartan subalgebra $\mathfrak{d}$.

The subalgebra $\mathfrak{d}$ is the Lie algebra of a subgroup D of type (C) of G . We may suppose that $C \subset D$. It remains to show that

$$H \supset C \iff \mathfrak{h} \supset \mathfrak{d}.$$

By XIII 5.5, the inclusion $\mathfrak{h} \supset \mathfrak{d}$ implies $H \supset D$, and *a fortiori* $H \supset C$. Conversely, by 5.1 the Lie algebra $\mathfrak{t}$ of C contains a regular element x of $\mathfrak{g}$. Hence $H \supset C$ implies $x \in \mathfrak{h}$, and, as observed above, this implies $\mathfrak{h} \supset \mathfrak{d}$. This completes the proof.

Let G be as in 5.2, and suppose that the infinitesimal rank of its fibers is locally constant, that is, condition (C_0) holds. By 5.2(c), there is a morphism of functors on $(\mathrm{Sch}/S)^\circ$,

$$\mathcal{C} \longrightarrow \mathcal{D},$$

where

$$\begin{aligned} \mathcal{C}(S') &= \{\text{Cartan subgroups of } G_{S'}\}, \\ \mathcal{D}(S') &= \{\text{subgroups of type (C) of } G_{S'}\}. \end{aligned}$$

By 3.10 and 5.2(a), the functor $\mathcal{D}$ is represented by a smooth quasi-projective prescheme over S . Let D be the universal subgroup of type (C) of $G_{\mathcal{D}}$. One may then define the functor

$$\mathcal{C}_D: (\mathrm{Sch}/\mathcal{D})^\circ \longrightarrow \mathrm{Ens}$$

for the $\mathcal{D}$ -group D , exactly as $\mathcal{C}$ was defined for the S -group G .

Proposition 5.5. *Under the preceding hypotheses, regard $\mathcal{C}$ as a functor over $\mathcal{D}$. Then $\mathcal{C}$ is $\mathcal{D}$ -isomorphic to the functor $\mathcal{C}_D$ of Cartan subgroups of D .*

This follows immediately from the next corollary.

Corollary 5.6. *Let G be a smooth S -prescheme in groups of finite presentation, and let D be a subgroup of type (C) of G . There is a bijective correspondence between the Cartan subgroups of G contained in D and the Cartan subgroups of D . More precisely, for a subgroup $H \subset D$, H is a Cartan subgroup of G if and only if it is a Cartan subgroup of D .*

This is a special case of XII 7.9(c), taking into account that, over an algebraically closed field, every subgroup of type (C) of G contains a Cartan subgroup of G .

For the next section, the principal result obtained here is 5.2(c) with H a Cartan subgroup of G . It permits the formulation of 5.5 and thus gives a useful dévissage of $\mathcal{C}$.

6 Applications to the Structure of Algebraic Groups

Theorem 6.1. *Let G be a smooth algebraic group over a field k . Consider the scheme $\mathcal{T}$ of maximal tori of G , isomorphic to the scheme $\mathcal{C}$ of Cartan subgroups of G . This is a homogeneous space under G^0 and a smooth affine connected algebraic scheme (XII 7.1(d)). Then $\mathcal{C}$ is a rational variety: its field of rational functions is a purely transcendental extension of k .*

We first prove the theorem when k is infinite. We may plainly suppose that G is connected, since $\mathcal{T}$, and hence $\mathcal{C}$, is unchanged when G is replaced by G^0 . Moreover, by XII 7.6, $\mathcal{C}$ is unchanged when G is divided by a central subgroup. We may therefore first divide by the center of G and suppose that G is affine (XII 6.1), and then divide by its reductive center (XII 4.1 and 4.4) and suppose that the reductive center of G is trivial (XII 4.7(b)).

We proceed by induction on $n = \dim G$, assuming the result for dimensions $n' < n$. If G is nilpotent, then $\mathcal{C}$ consists of a single k -rational point and the assertion is trivial. Otherwise, the Lie algebra of G is not nilpotent (1.3). Its Cartan subalgebras, and therefore the subgroups of type (C) of G , have some dimension $n' < n$. Consider the morphism

$$\mathcal{C} \longrightarrow \mathcal{D}$$

of 5.5. By 3.10, since the infinite-field Lie algebra $\mathfrak{g}$ contains a regular element, $\mathcal{D}$ is a rational variety. Thus its function field K is purely transcendental over k .

Let x be the generic point of $\mathcal{D}$. By 5.5, the fiber of $\mathcal{C}$ at x is the scheme of Cartan subgroups of a smooth connected algebraic group D_x over $K = \kappa(x)$, namely the generic subgroup of type (C) of G . The function field L of $\mathcal{C}$ is therefore the function field of $\mathcal{C}_{D_x}$. Since $\dim D_x = n' < n$, the induction hypothesis makes L purely transcendental over K , and hence, by transitivity, purely transcendental over k .

Suppose now that k is finite. A different argument is required. We may again suppose that G is affine and connected. Since k is perfect, the radical of $G_{\bar{k}}$ descends to a subgroup R of G . Suppose first that $R \neq G$, so that G is not solvable, and let

$$u: G \longrightarrow G' = G/R$$

be the canonical morphism. By XII 7.1(e), it induces

$$v: \mathcal{C}_G \longrightarrow \mathcal{C}_{G'}, \quad C \longmapsto u(C).$$

Let x be the generic point of $\mathcal{C}_{G'}$, and put $K = \kappa(x)$. The fiber $v^{-1}(x)$ is the scheme of Cartan subgroups of G_K whose image in G'_K is the generic Cartan subgroup C'_x of G' . By XII 7.9(c), it is also the scheme of Cartan subgroups of

$$H = u_K^{-1}(C'_x).$$

The field K is infinite. The infinite-field case just proved therefore shows that the function field L of $\mathcal{C}_G$, which is also that of $\mathcal{C}_H$, is purely transcendental over K .¹²

It remains to prove that K is purely transcendental over k , and we are thus reduced to the case in which G is semisimple. Dividing by its reductive center, we may further suppose that G is adjoint. Then 3.18 gives $\mathcal{C}_G \simeq \mathcal{D}_G$, and by 3.10 it is enough to show that $\mathfrak{g}$ has a regular point. As noted in 3.20, this is an unpublished result of Chevalley.¹³

It remains only to treat the case in which k is finite and G is connected, affine, and solvable. In fact, the next result holds over an arbitrary field.

¹²The French source refers here to the “already proved part of 7.1”; the argument is the infinite-field part of Theorem 6.1.

¹³See the Appendix by J.-P. Serre.

Corollary 6.2. *Let G be a smooth solvable algebraic group over a field k . Then the variety $\mathcal{C}$ of Cartan subgroups of G is isomorphic to an affine space $\operatorname{Spec} k[t_1, \dots, t_N]$.*

We may suppose that G is connected and affine. Let G_∞ be the terminal group in the descending central series

$$G_0 = G, \quad G_{i+1} = [G, G_i].$$

Thus G_∞ is the smallest invariant algebraic subgroup of G for which G/G_∞ is nilpotent. Let C be a Cartan subgroup of G , which exists by 1.1. Its image in G/G_∞ is a Cartan subgroup and hence is all of G/G_∞ . Consequently the morphism

$$G_\infty \longrightarrow \mathcal{C}, \quad g \longmapsto \operatorname{Ad}(g)C$$

is an epimorphism and identifies

$$\mathcal{C} \simeq G_\infty / (N \cap G_\infty),$$

where N is the normalizer of C in G . In fact $N = C$, as recalled in 4.4, but that is immaterial here.

The group $U = G_\infty$ is smooth, connected, and unipotent: over $\bar{k}$ it lies in the unipotent part of G , by the known structure of smooth affine solvable groups *Bible* 6, Theorem 3. If k is perfect, as in the finite-field case needed for 6.1, it follows easily that U is k -unipotent: it has a composition series by algebraic subgroups U_i with $U_i/U_{i+1} \simeq \mathbb{G}_a$.

Rosenlicht proved that a group of the form G_∞ above is k -unipotent without any restriction on k (M. Rosenlicht, *Questions of rationality for solvable algebraic groups over non perfect fields*, *Annali di Matematica*, 1963, pp. 97–120, Theorem 4, Corollary 2). We admit this substantially more delicate result and apply the following lemma.

Lemma 6.3. *Let U be a smooth connected algebraic group over a field k , and let $X = U/V$ be a homogeneous space under U having a k -rational point. If U is k -unipotent, then, as a k -scheme, X is isomorphic to an affine space $\operatorname{Spec} k[t_1, \dots, t_N]$.*

Choose a composition series

$$U = U_0 \supset U_1 \supset \dots \supset U_n = (e)$$

by smooth connected normal subgroups of U , with $U_i/U_{i+1} \simeq \mathbb{G}_a$. The groups

$$K_i = U_i V$$

are algebraic subgroups of U . They need not be smooth or connected when V is not, but this does not matter, and K_{i+1} is normal in K_i . The canonical epimorphism

$$U_i/U_{i+1} \longrightarrow K_i/K_{i+1}$$

shows that K_i/K_{i+1} is either the unit group or isomorphic to $\mathbb{G}_a/H_i$, where H_i is a finite subgroup of $\mathbb{G}_a$. Rosenlicht's Theorem 2 in the cited paper gives

$$K_i/K_{i+1} \simeq \mathbb{G}_a;$$

this last assertion is immediate when k is perfect.

Put $X_i = U/K_i$. We prove inductively that every X_i is an affine space. If K_i/K_{i+1} is trivial, then $X_i = X_{i+1}$. Otherwise X_{i+1} is a principal $\mathbb{G}_a$ -bundle over X_i . Since X_i is affine, the bundle is trivial, so

$$X_{i+1} \simeq X_i \times \mathbb{G}_a.$$

The induction proves the lemma. It proves 6.2 and thereby completes the proof of 6.1.

Corollary 6.4. *Let G be a smooth algebraic group over an infinite field k . Then the set of k -rational points of $\mathcal{C}$, in the notation of 6.1, is Zariski dense. The union of the Cartan subgroups of G is dense in G .*

The first assertion holds for every rational variety over an infinite field and is, for us, the most important arithmetic consequence of the rationality result. The second follows from the first and the density theorem XIII 2.1.

Corollary 6.5. *Let G be a smooth connected algebraic group over k . Then the variety Z of regular semisimple points of G (XIII 3.5) is unirational. In particular, if k is infinite, its k -rational points are dense in Z .*

Indeed, Z is an open subscheme of a torus over $\mathcal{C}$. Its function field L is therefore the function field of a torus over the function field K of $\mathcal{C}$, namely the generic maximal torus of G . Thus L is unirational over K by XIII 3.4. Since 6.1 makes K purely transcendental over k , L is unirational over k .

Corollary 6.6. *Let G be a smooth connected algebraic group over k , and let H be the subgroup generated by the subvariety Z of regular semisimple points. Equivalently (XII 8.2), H is the smallest invariant algebraic subgroup of G for which G/H has reductive rank zero; over $\bar{k}$, G/H is an extension of an abelian variety by a smooth connected unipotent group. Then H is unirational.*

In particular, suppose $G = H$; equivalently (XII 8.4), suppose G is affine and there is no nontrivial homomorphism $G_{\bar{k}} \rightarrow \mathbb{G}_a$. Then G is unirational. If, in addition, k is infinite, $G(k)$ is dense in G .

This follows immediately from 6.5. Indeed, if Z_i are smooth connected unirational k -preschemes and $u_i: Z_i \rightarrow G$ are morphisms, then the algebraic subgroup of G generated by the u_i is unirational (VI_B 7.1).

An interesting special case of either 6.5 or 6.6 is the following.

Corollary 6.7. *Let G be a smooth connected affine algebraic group of unipotent rank zero. Then G is unirational.*

The conclusion of 6.4 may be sharpened as follows.

Corollary 6.8. *Let G be a smooth connected algebraic group over an infinite field k , and let C be a Cartan subgroup of G . Then the union of the conjugates of C by regular semisimple elements of $G(k)$ is dense in G .*

This follows at once from 6.4 and XIII 3.6, which says that the morphism

$$\varphi: Z \times C \longrightarrow G, \quad \varphi(t, c) = \text{Ad}(t)c,$$

is dominant. The same result also implies the next corollary, without any assumption that k is infinite.

Corollary 6.9. *Let G be a smooth connected algebraic group over k , and let H be a smooth connected algebraic subgroup having the same reductive rank and the same nilpotent rank as G . Equivalently, over $\bar{k}$, the group $H_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$. If H is unirational, so is G . If $H(k)$ is dense in H , then $G(k)$ is dense in G .*

The morphism

$$\varphi: Z \times H \longrightarrow G, \quad \varphi(t, h) = \text{Ad}(t)h,$$

is dominant. By 6.5, Z is unirational, and by hypothesis so is H , hence so are $Z \times H$ and G . The density assertion is proved in the same way.

We recover the following well-known result, due to Chevalley in characteristic zero and to Rosenlicht in characteristic $p > 0$.

Corollary 6.10. *Let G be a smooth connected affine algebraic group over a perfect field k . Then G is unirational. If k is also infinite, $G(k)$ is dense in G .*

By 1.1, G has a Cartan subgroup C . By 6.9, it is enough to prove that C is unirational. Since k is perfect, Galois descent from the algebraically closed case *Bible* 6, Theorem 2 gives

$$C = T \times C_u,$$

where T is the maximal torus of C and C_u is smooth, connected, and unipotent. The torus T is unirational by XIII 3.4. Moreover, C_u is k -unipotent because k is perfect, so 6.3 applies to C_u .

Remarks 6.11. (a) Rosenlicht gave examples of twisted forms of $\mathbb{G}_a$ over a nonperfect field having only finitely many rational points; *a fortiori*, they are not unirational. Chevalley, on the other hand, gave an example of a torus over a field of characteristic zero that is not rational. By contrast, Chevalley's theory of semisimple groups shows that, over an algebraically closed field, every smooth connected affine algebraic group is rational. In any case, the question of unirationality arises only for affine algebraic groups: by Chevalley's structure theorem, a unirational algebraic group is necessarily affine.

(b) With the notation of 6.6, it is tempting to seek a criterion for the unirationality of G in terms of G/H , which is unipotent when G is affine. Clearly G/H must be unirational; is that condition also sufficient? An example of Rosenlicht shows that a smooth connected unipotent algebraic group U may be rational without being k -unipotent.

(c) Over a finite field k , it would be interesting to study questions of “density” such as the following one, raised by Rosenlicht: if G is a smooth connected algebraic group over k , is G generated by its Cartan subgroups?¹⁴

Editors' note. The editors did not identify this result in Steinberg's *Collected Papers*; here is a proof based on the Bruhat decomposition.

Let G/k be semisimple over the finite field k , and let $G^\sharp$ be the subgroup generated by its k -subtori. The question is stable under central isogeny and restriction of scalars, so, as in Lemma 1 of the Appendix below, it is enough to treat the geometrically simple case. The group G is quasi-split by Lang's theorem and admits a Killing pair (T, B) . By definition $T \subset G^\sharp$, hence T normalizes $G^\sharp$. The big cell

$$R_u(B^-)T R_u(B)$$

shows that it suffices to prove $R_u(B) \subset G^\sharp$. Let S be the maximal split k -torus of T . For the relative root system $\Phi(G, S)$, choose a basis Δ_k , and denote by $U(\alpha)$ the unipotent subgroup associated with α (A. Borel, *Linear Algebraic Groups*, second edition, 1991, Proposition 21.9). Since $R_u(B)$ is generated by the $U(\alpha)$, $\alpha \in \Delta_k$, it is enough to show $U(\alpha) \subset G^\sharp$.

The classification supplies a quasi-split semisimple subgroup G_α of type A_1 , 2A_1 , 3A_1 , or 2A_2 , with

$$U(\alpha) \subset G_\alpha \subset G.$$

Thus it is enough to consider $G = \mathrm{PGL}_2$ or $G = \mathrm{SU}_3(K)$, where K/k is the unique quadratic extension.

¹⁴This question was subsequently answered in the affirmative by Steinberg.

For PGL_2 , since $G^\sharp$ contains the standard split torus T , the possibilities up to $G(k)$ -conjugacy are

$$G^\sharp = T, \quad G^\sharp = B, \quad G^\sharp = \mathrm{PGL}_2.$$

The first is excluded because $G^\sharp$ contains the k -torus $R_{K/k}(\mathbb{G}_m)/\mathbb{G}_m$. The preceding discussion also shows that $G^\sharp = B$ forces $G^\sharp = \mathrm{PGL}_2$. Hence $G^\sharp = \mathrm{PGL}_2$.

For $G = \mathrm{SU}_3(K)$, one has $\Delta_k = \{\alpha\}$, and the possibilities up to conjugacy are

$$G^\sharp = T, \quad B, \quad U(2\alpha) \rtimes T, \\ \langle U(2\alpha), U(-2\alpha), T \rangle = \mathrm{SL}_2 \cdot T, \quad \text{or } G.$$

The case B is excluded as for PGL_2 . The case $\mathrm{SL}_2 \cdot T$ is excluded because $G^\sharp$ contains the k -torus

$$R_{K/k}^1 \left(R_{K_3/K}^1(\mathbb{G}_m) \right),$$

where K_3/K is the degree-three extension, and the isogeny class of this torus is irreducible. The same torus excludes T and $U(2\alpha) \rtimes T$. Hence $G^\sharp = G$.

For Rosenlicht's question one may reduce to the affine case by dividing by the center. In the semisimple case, an affirmative answer would also follow from a strengthening of Chevalley's existence result for regular points mentioned in 3.20: one would need a regular element of $\mathfrak{g}$ outside $\mathfrak{h} = \mathrm{Lie}(H)$, where $H \neq G$ is a prescribed smooth algebraic subgroup of an adjoint semisimple group G over the finite field k .

7 Appendix: Existence of Regular Elements over Finite Fields

by J.-P. Serre

Throughout the Appendix, k denotes a finite field, $\bar{k}$ its algebraic closure, and $\mathcal{G} = \mathrm{Gal}(\bar{k}/k)$. If $q = \mathrm{Card}(k)$, then $\mathcal{G}$ is topologically generated by the Frobenius element

$$F: x \mapsto x^q.$$

We shall prove the following theorem.¹⁵

Theorem. *Let G be an adjoint semisimple group defined over k , and let $\mathfrak{g}$ be its Lie algebra. Then the k -Lie algebra $\mathfrak{g}(k)$ contains a regular element.*

Remarks. (1) Recall that “adjoint” means that the center of G , as a subgroup scheme, is trivial. In view of the Chevalley seminar, this also means that, for a maximal torus T of G , the character group $X(T)$ of $T_{\bar{k}}$ is generated by the set R of roots. In particular, the rank of $\mathfrak{g}$ equals $\dim T$, the rank of G .

(2) The editor does not know whether the adjoint hypothesis is indispensable.

¹⁵The theorem is due to Chevalley. The editor wishes to express his gratitude to American Express which, by misplacing a trunk of Chevalley's manuscripts, obliged him to reconstruct the proof.

Lemma 1. *It is enough to prove the theorem when G is geometrically simple.*

Here geometrically simple means that $G_{\bar{k}}$ has no smooth normal subgroup scheme other than $G_{\bar{k}}$ and (e) ; equivalently, the associated root system is irreducible.

We may first suppose that G is indecomposable over k . Then $G_{\bar{k}}$ is a product of geometrically simple components S , permuted transitively by $\mathcal{G}$. If $\mathcal{H} \subset \mathcal{G}$ fixes one component S , and K is its fixed field, then S is defined over K , and the standard restriction-of-scalars argument gives

$$G = R_{K/k}(S).$$

Similarly,

$$\mathfrak{g} = R_{K/k}(\mathfrak{s}),$$

where $\mathfrak{s}$ is the Lie algebra of S . If the theorem holds for S , there is an element

$$x \in \mathfrak{s}(K) = \mathfrak{g}(k)$$

regular in $\mathfrak{s}$, and one checks immediately that it is regular in $\mathfrak{g}$. This proves the lemma.

Henceforth assume that G is geometrically simple, and choose a maximal torus T of G . Let X be the character group of $T_{\bar{k}}$, let R be its root system, W its Weyl group, and E the group of automorphisms of X preserving R . Then W is normal in E .

For another maximal torus T' , write X', R', W', E' for the corresponding objects. If $y \in G(\bar{k})$ satisfies

$$yT_{\bar{k}}y^{-1} = T'_{\bar{k}},$$

then $\text{int}(y)$ identifies (X', R', W', E') with (X, R, W, E) . Replacing y changes this identification by an element of W .

Frobenius acts on X' while preserving R' , and thus defines an element $f_{T'} \in E'$. Put $f = f_T$. After the above identification $E' \simeq E$, the element $f_{T'}$ becomes an element $f' \in E$, well defined up to W -conjugacy. We now compare f' with f .

Lemma 2. *One has $f' \equiv f \pmod{W}$. Conversely, every $f' \in E$ satisfying this condition arises from a maximal torus T' of G .*

Put $\bar{T} = T_{\bar{k}}$ and $\bar{T}' = T'_{\bar{k}}$, and choose $y \in G(\bar{k})$ such that

$$y\bar{T}y^{-1} = \bar{T}'.$$

Since $y^q\bar{T}y^{-q} = \bar{T}'$, the element

$$n = y^{-1}y^q$$

belongs to the normalizer $N(T)(\bar{k})$. The action of f' on the points of $\bar{T}$ is

$$\begin{aligned} f'(t) &= y^{-1}(yty^{-1})^q y \\ &= y^{-1}y^q t^q y^{-q} y = nt^q n^{-1}. \end{aligned}$$

If $w \in W$ is represented by n , then f' and $f \circ w$ have the same action on $\bar{T}(\bar{k})$, hence on its characters. Thus $f' = f \circ w$, and $f' \equiv f \pmod{W}$.

Conversely, represent a given $w \in W$ by $n \in N(T)(\bar{k})$. Lang's theorem gives

$$n = y^{-1}y^q$$

for some $y \in G(\bar{k})$. Then $y\bar{T}y^{-1}$ is defined over k , and the preceding calculation shows that its corresponding element is $f \circ w$.

Lemma 3. *Let X, R, W, E be as above, with R irreducible, and let $\varphi \in E/W$. There exist a representative $f \in E$ of φ and roots $\theta_1, \dots, \theta_n$ such that:*

1. $(\theta_1, \dots, \theta_n)$ is a basis of X ;
2. R is the union of the orbits of the θ_i under the powers of f : every $a \in R$ has the form $a = f^m \theta_i$ for suitable m and i .

The proof will be given below. We first need a lemma of linear algebra.

Lemma 4. *Let V be a vector space over k , and let $(\theta_1, \dots, \theta_n)$ be a basis of the dual of $V \otimes_k \bar{k}$. Then there is an $x \in V$ such that $\theta_i(x) \neq 0$ for every i .*

Let V^* be the dual of V , and let W_i be the $\bar{k}$ -subspace of $V^* \otimes_k \bar{k}$ generated by the Galois conjugates of θ_i . It descends to a k -subspace, again denoted W_i . The natural map

$$W_1 \otimes \dots \otimes W_n \longrightarrow \bigwedge^n V^*$$

is nonzero: after extension to $\bar{k}$, it contains $\theta_1 \wedge \dots \wedge \theta_n \neq 0$. Hence V^* has a basis $u_1, \dots, u_n$ with $u_i \in W_i$. Choose $x \in V$ with $u_i(x) \neq 0$ for all i , for example with $u_i(x) = 1$. If $\theta_i(x) = 0$, every Galois conjugate of θ_i vanishes at x , and so does u_i , a contradiction.

End of the proof of the theorem. By Lemmas 2 and 3, choose a maximal torus T whose element f has the properties of Lemma 3. Let Y be the dual of X . The Lie algebra of T satisfies

$$\mathfrak{t}(\bar{k}) \simeq Y \otimes_{\mathbb{Z}} \bar{k},$$

compatibly with the Galois action. Put $V = \mathfrak{t}(k)$. An element $x \in V$ is regular precisely when it is annihilated by no root $\alpha \in R$, viewed as a linear form in $V^* \otimes_k \bar{k}$.

Lemma 4 supplies $x \in V$ annihilated by none of the θ_i . Every root is a Galois conjugate of one of the θ_i , by property (2) of Lemma 3. Hence no root annihilates x , and x is regular. This proves the theorem.

Proof of Lemma 3. The proof uses Coxeter transformations. Let $a_1, \dots, a_n$ be a simple system of roots of R , and let r_i be the reflection corresponding to a_i . Put

$$c = r_1 \cdots r_n.$$

Then $c \in W$. Although c depends on the simple system and on its ordering, its conjugacy class does not. It is called the Coxeter element. We shall use, without proof, the fact that 1 is not an eigenvalue of c .

Lemma 5. Put

$$\theta_i = r_n \cdots r_{i+1}(a_i).$$

Then:

- (a) the θ_i form a basis of the group X generated by the roots;
- (b) $\theta_i > 0$ and $c(\theta_i) < 0$ for every i ;
- (c) every root $a \in R$ such that $a > 0$ and $c(a) < 0$ is one of the θ_i ;
- (d) R is the union of the orbits of the θ_i under the powers of c .

Indeed,

$$\theta_i = a_i + \sum_{j>i} m_{ij} a_j, \quad m_{ij} \in \mathbb{Z},$$

which proves (a). Assertions (b) and (c) follow from the observation that r_i preserves the sign of every root other than $\pm a_i$, and reverses the sign of $\pm a_i$.

For (d), an orbit of c cannot consist entirely of positive roots, nor entirely of negative roots. Otherwise the sum of the roots in that orbit would give a nonzero element of X fixed by c , contrary to the fact that 1 is not an eigenvalue. Hence every orbit contains a root $a > 0$ for which $c(a) < 0$, and (c) applies.

Remark. The proof was recalled only for the reader's convenience; one could instead refer to the standard texts on Coxeter transformations, for example Koszul, Séminaire Bourbaki 1959/1960, Exposé 191. They also show that all orbits of c have the same number h of elements and each contains exactly one θ_i . In particular,

$$h = \frac{\text{Card}(R)}{n}.$$

We return to Lemma 3 and distinguish three cases.

(1) Suppose that $\varphi \in E/W$ is the identity. We must choose $f \in W$; take $f = c$. Lemma 5 shows that this works.

(2) Suppose that R has type A_n , where $n \geq 2$ is even, and that φ is the unique nontrivial element of E/W . Then $-1 \notin W$, and we take

$$f = -c.$$

The Coxeter element c has order $h = n + 1$, which is odd. If $a \in R$, Lemma 5 gives $a = c^m \theta_i$ for suitable m, i . Adding h to m if necessary, we may suppose that m is even. Then

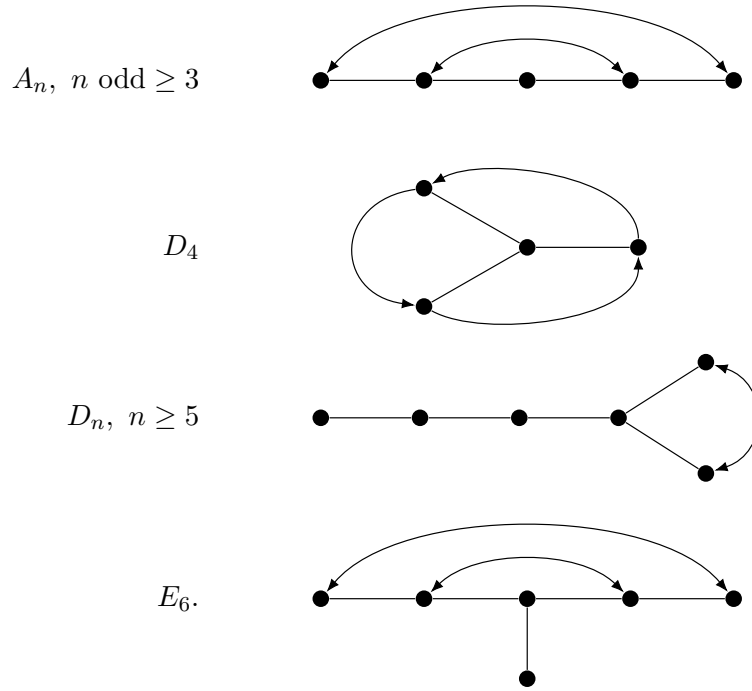
$$a = (-c)^m \theta_i = f^m \theta_i.$$

Thus the orbits of the θ_i under f fill R .

(3) Suppose that $\varphi \in E/W$ is nontrivial and that R has one of the following types:

$$A_n \ (n \geq 3 \text{ odd}), \quad D_4, \quad D_n \ (n \geq 5), \quad E_6.$$

These, together with the even A_n case treated separately above, are exactly the cases in which E/W is nontrivial. The following diagrams record the corresponding Dynkin-diagram automorphisms.



Let S be a simple system of roots, not yet ordered. The group E is the semidirect product of W with the group Ψ of permutations of S preserving the Cartan matrix, or equivalently the Dynkin diagram. Thus $E/W \simeq \Psi$, and φ corresponds to some $\psi \in \Psi$.

Inspection of the diagrams shows that each ψ -orbit in S consists of pairwise orthogonal roots. This fails for A_n with n even, which is why that case was separated. If σ is such an orbit, the reflections r_a , $a \in \sigma$, commute; write

$$\rho_\sigma = \prod_{a \in \sigma} r_a.$$

Plainly ρ_σ commutes with ψ .

Choose a total order on S for which every ψ -orbit is a segment, and write $S = \{a_1, \dots, a_n\}$. Let $\theta_1, \dots, \theta_n$ and $c = r_1 \cdots r_n$ be defined as above. The element c is a product of the ρ_σ , with the orbits in some order, so it commutes with ψ . Put

$$f = c\psi.$$

The automorphism ψ permutes the θ_i . Indeed, $\psi(\theta_i) > 0$, and, since c commutes with ψ ,

$$c(\psi(\theta_i)) = \psi(c(\theta_i)) < 0.$$

Lemma 5(c) therefore gives $\psi(\theta_i) = \theta_j$ for some j .

Finally, if $a \in R$, Lemma 5 gives $a = c^m \theta_i$ for some m, i . Hence

$$a = f^m(\psi^{-m} \theta_i) = f^m \theta_j$$

for a suitable j . Thus f and the θ_i satisfy both conditions of Lemma 3, completing its proof.

Remark. Except in case (2), one can prove that every orbit of f has exactly

$$h = \frac{\text{Card}(R)}{n}$$

elements and contains exactly one θ_i . In case (2), some of the θ_i are superfluous.